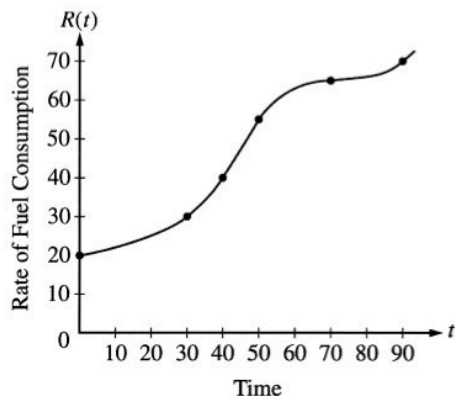


2-1-1

1.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

Use data from the table to find an approximation for $R'(45)$. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min² with correct units



0	1	2
---	---	---

The student response earns two of the following points:

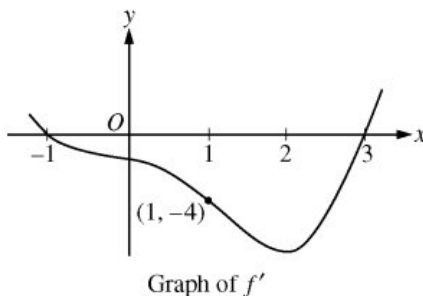
1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min² with correct units

2-1-1

2.



Let f be a twice-differentiable function defined on the interval $-1.2 < x < 3.2$ with $f(1) = 2$. The graph of f' , the derivative of f , is shown above. The graph of f' crosses the x -axis at $x = -1$ and $x = 3$ and has a horizontal tangent at $x = 2$. Let g be the function given by $g(x) = e^{f(x)}$.

- Write an equation for the line tangent to the graph of g at $x = 1$.
- For $-1.2 < x < 3.2$, find all values of x at which g has a local maximum. Justify your answer.
- The second derivative of g is $g''(x) = e^{f(x)} [(f'(x))^2 + f''(x)]$. Is $g''(-1)$ positive, negative, or zero? Justify your answer.
- Find the average rate of change of g' , the derivative of g , over the interval $[1, 3]$.

Part A

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

2-1-1

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.

Part B

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So, g' changes from positive to negative only when f' changes from positive to negative. This occurs at $x = -1$ only. Thus, g has a local maximum at $x = -1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So, g' changes from positive to negative only when f' changes from positive to negative. This occurs at $x = -1$ only. Thus, g has a local maximum at $x = -1$.

Part C

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[(f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.

2-1-1



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[(f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.

Part D

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{e^{f(3)}f'(3) - e^{f(1)}f'(1)}{2} = 2e^2$$



0	1	2
---	---	---

The student response earns all of the following points:

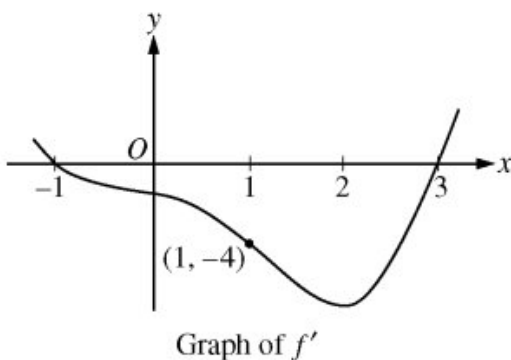
1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{e^{f(3)}f'(3) - e^{f(1)}f'(1)}{2} = 2e^2$$

2-1-1

3.



Let f be a twice-differentiable function defined on the interval $-1.2 < x < 3.2$ with $f(1)=2$. The graph of f' , the derivative of f , is shown above. The graph of f' crosses the x -axis at $x=-1$ and $x=3$ and has a horizontal tangent at $x=2$. Let g be the function given by $g(x)=e^{f(x)}$.

Find the average rate of change of g' , the derivative of g , over the interval $[1, 3]$.

Part D

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g'(3)-g'(1)}{3-1} = \frac{e^{f(3)}f'(3)-e^{f(1)}f'(1)}{2} = 2e^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g'(3)-g'(1)}{3-1} = \frac{e^{f(3)}f'(3)-e^{f(1)}f'(1)}{2} = 2e^2$$

2-1-1

4. Let f be a twice-differentiable function such that $f(2) = 5$ and $f(5) = 2$. Let g be the function given by $g(x) = f(f(x))$.
- (a) Explain why there must be a value c for $2 < c < 5$ such that $f'(c) = -1$.
- (b) Show that $g'(2) = g'(5)$. Use this result to explain why there must be a value k for $2 < k < 5$ such that $g''(k) = 0$.
- (c) Show that if $f''(x) = 0$ for all x , then the graph of g does not have a point of inflection.
- (d) Let $h(x) = f(x) - x$. Explain why there must be a value r for $2 < r < 5$ such that $h(r) = 0$.

Part A

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$

Part B

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

2-1-1

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on $[2, 5]$ guarantees there is

a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on $[2, 5]$ guarantees there is

a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.

Part C

1 point is earned if considers g''

1 point is earned for $g''(x) = 0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$$

If $f''(x) = 0$ for all x , then

$$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0 \text{ for all } x.$$

2-1-1

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points. OR

If $f''(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers g''

1 point is earned for $g''(x) = 0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$$

If $f''(x) = 0$ for all x , then

$$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points. OR

If $f''(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.

Part D

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

$$\text{Let } h(x) = f(x) - x.$$

$$h(2) = f(2) - 2 = 5 - 2 = 3$$

$$h(5) = f(5) - 5 = 2 - 5 = -3$$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2 < r < 5$, such that $h(r) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

$$\text{Let } h(x) = f(x) - x.$$

$$h(2) = f(2) - 2 = 5 - 2 = 3$$

2-1-1

$$h(5) = f(5) - 5 = 2 - 5 = -3$$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2 < r < 5$, such that $h(r) = 0$.

5. Let f be a twice-differentiable function such that $f(2)=5$ and $f(5)=2$. Let g be the function given by $g(x)=f(f(x))$. Explain why there must be a value c for $2 < c < 5$ such that $f'(c)=-1$.

Part A

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$

2-1-1

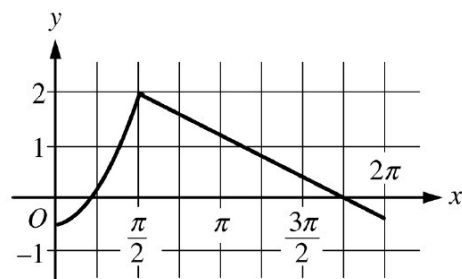
6. Let f be the function defined by $f(x) = e^x \cos(x)$.

(a) Find the average rate of change of f on the interval $0 \leq x \leq \pi$.

(b) What is the slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$?

(c) Find the absolute minimum value of f on the interval $0 \leq x \leq 2\pi$. Justify your answer.

(d) Let g be a differentiable function such that $g\left(\frac{\pi}{2}\right) = 0$. The graph of g' , the derivative of g , is shown below. Find the value of $\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)}$ or state that it does not exist. Justify your answer.



Graph of g'

Part A

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.



0

1

The student response earns one of the following points:

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.

Part B

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.

2-1-1



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.

Part C

1 point is earned for sets $f'(x) = 0$

1 point is earned for identifies $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ as candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.

2-1-1



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for sets $f'(x) = 0$

1 point is earned for identifies as $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.

Part D

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

2-1-1

$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$

2-1-1

7. Let f be the function defined by $f(x) = e^x \cos(x)$.
Find the average rate of change of f on the interval $0 \leq x \leq \pi$.

Part A

1 point is earned for: answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is

$$\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}.$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for: answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is

$$\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}.$$

2-1-1

8. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \sqrt{9-x^2} & \text{for } -3 \leq x \leq 0 \\ -x + 3 \cos\left(\frac{\pi x}{2}\right) & \text{for } 0 < x \leq 4 \end{cases}$$

Let f be the function defined above.

- (a) Find the average rate of change of f on the interval $-3 \leq x \leq 4$.
- (b) Write an equation for the line tangent to the graph of f at $x = 3$.
- (c) Find the average value of f on the interval $-3 \leq x \leq 4$.
- (d) Must there be a value of x at which $f(x)$ attains an absolute maximum on the closed interval $-3 \leq x \leq 4$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point does not require a simplified answer. Substitution of function values is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ answer

Solution:

$$\text{Average rate of change} = \frac{f(4) - f(-3)}{4 - (-3)} = \frac{-1 - 0}{7} = -\frac{1}{7}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct and labeled value of $f'(3)$.

The second point can be earned with a consistent tangent line equation created from an incorrect value of $f'(3)$.



0	1	2
---	---	---

2-1-1

The student response accurately includes **both** of the criteria below.

- ☐ $f'(3)$
- ☐ equation

Solution:

$$f(3) = -3 + 3 \cos\left(\frac{3\pi}{2}\right) = -3$$

$$\text{For } 0 < x < 4, f'(x) = -1 + \left(-3 \sin\left(\frac{\pi x}{2}\right)\right) \cdot \frac{\pi}{2}$$

$$f'(3) = -1 + \left(-3 \sin\left(\frac{3\pi}{2}\right)\right) \cdot \frac{\pi}{2} = -1 + \frac{3\pi}{2}$$

$$\text{An equation for the tangent line is } y = -3 + \left(-1 + \frac{3\pi}{2}\right)(x - 3).$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The constant $\frac{1}{b-a}$ is part of the fourth point.

Simplification is not required for the second point.

The third point is earned with the antiderivative $\frac{-x^2}{2} + \frac{6}{\pi} \sin\left(\frac{\pi x}{2}\right)$. The third point can also be earned with $\frac{-x^2}{2} + \frac{2}{\pi} \sin\left(\frac{\pi x}{2}\right)$, though this response will not be eligible for the fourth point.

Simplification is not required for the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ integrals of f over $-3 \leq x \leq 0$ and $0 \leq x \leq 4$
- ☐ value of $\int_{-3}^0 \sqrt{9-x^2} \, dx$
- ☐ antiderivative of $-x + 3 \cos\left(\frac{\pi x}{2}\right)$
- ☐ answer

Solution:

The average value of f on the interval $-3 \leq x \leq 4$ is $\frac{1}{4-(-3)} \int_{-3}^4 f(x) \, dx$.

$$\int_{-3}^4 f(x) \, dx = \int_{-3}^0 f(x) \, dx + \int_0^4 f(x) \, dx$$

$$\int_{-3}^0 f(x) \, dx = \int_{-3}^0 \sqrt{9-x^2} \, dx = \frac{9\pi}{4}$$

2-1-1

$$\int_0^4 f(x) \, dx = \int_0^4 \left(-x + 3 \cos \left(\frac{\pi x}{2} \right) \right) \, dx = \left[-\frac{1}{2}x^2 + \frac{6}{\pi} \sin \left(\frac{\pi x}{2} \right) \right]_0^4 = -8$$

$$\frac{1}{4 - (-3)} \int_{-3}^4 f(x) \, dx = \frac{1}{7} \left(\frac{9\pi}{4} - 8 \right)$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned by evaluating each branch of the function at $x = 0$ instead of evaluating the one-sided limits, but the response will not be eligible for the second point.

The second point is earned with correct limit notation for the first point, clear communication that

$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) = 3$, and a statement that the function is continuous on the closed interval, and therefore must have an absolute maximum. Specifically naming the Extreme Value Theorem is not required to earn this point. Responses that indicate that the function is differentiable on the interval $[-3, 4]$ will not earn the point.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ continuity at $x = 0$
- ☐ answer with justification

Solution:

$\lim_{x \rightarrow 0^-} f(x) = f(0) = 3$ and $\lim_{x \rightarrow 0^+} f(x) = 3$, so f is continuous at $x = 0$.

Because f is continuous on $[-3, 4]$, the Extreme Value Theorem guarantees that f attains an absolute maximum on $[-3, 4]$.

2-1-1

9.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	6
$f(x)$	-5	4	1	7
$f'(x)$	-1	2	-2	4

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(x^2 - x)$.

- (a) What is the value of $g'(3)$?
- (b) It is known that $g''(0) = -1$. What is the value of $f''(0)$?
- (c) Is there a value c , for $0 < c < 3$, such that $g(c) = 5$? Justify your answer.
- (d) Let h be the function with derivative given by $h'(x) = 4e^{\cos x}$. At what value of x in the interval $-3 \leq x \leq 0$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $-3 \leq x \leq 0$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes one of the criteria below.

- ☐ $g'(x)$
- ☐ $g'(3)$

Solution:

2-1-1

$$g'(x) = f'(x^2 - x)(2x - 1)$$

$$g'(3) = f'(6)(5) = 4(5) = 20$$

Part B

Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $g''(x)$ (2 points for full credit)
- ☐ answer

Solution:

$$g''(x) = f''(x^2 - x)(2x - 1)^2 + f'(x^2 - x)(2)$$

$$g''(0) = f''(0)(-1)^2 + f'(0)(2)$$

$$-1 = f''(0) + 2(2) \Rightarrow f''(0) = -5$$

Part C

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ g is continuous, with reason
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = 4 \text{ and}$$

2-1-1

$$g(3) = f(6) = 7.$$

$$\Rightarrow g(0) < 5 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 5$.

Part D

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ average rate of change

☐ answer

Solution:

The average rate of change of f over the interval $-3 \leq x \leq 0$

$$\frac{f(0) - f(-3)}{0 - (-3)} = \frac{4 - (-5)}{3} = 3.$$

$$h'(x) = 4e^{\cos x} = 3 \text{ for } -3 \leq x \leq 0. \Rightarrow x = -1.862602$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $-3 \leq x \leq 0$ at $x = -1.863$ (or -1.862).

2-1-1

10.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	4
$f(x)$	5	-1	2	7
$f'(x)$	-2	4	0	1

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(2x - x^2)$.

- What is the value of $g'(-1)$?
- It is known that $g''(0) = 0$. What is the value of $f''(0)$?
- Is there a value c , for $0 < c < 3$, such that $g(c) = 2$? Justify your answer.
- Let h be the function with first derivative given by $h'(x) = 4xe^x$. At what value of x in the interval $0 \leq x \leq 4$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $0 \leq x \leq 4$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ $g'(-1)$

Solution:

2-1-1

$$g'(x) = f'(2x - x^2)(2 - 2x)$$

$$g'(-1) = f'(-3)(4) = (-2)(4) = -8$$

Part B

Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $g''(x)$ (2 points for full credit)
- ☐ answer

Solution:

$$g''(x) = f''(2x - x^2)(2 - 2x)^2 + f'(2x - x^2)(-2)$$

$$g''(0) = f''(0)(2)^2 + f'(0)(-2)$$

$$0 = 4f''(0) + 4(-2) \Rightarrow f''(0) = 2$$

Part C

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ g is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = -1 \text{ and } g(3) = f(-3) = 5.$$

2-1-1

$$\Rightarrow g(0) < 2 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 2$.

Part D

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

The average rate of change of f over the interval $0 \leq x \leq 4$ is $\frac{f(4)-f(0)}{4-0} = \frac{7-(-1)}{4} = 2$.

$$h'(x) = 4xe^x = 2 \text{ for } 0 \leq x \leq 4. \Rightarrow x = 0.351734$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $0 \leq x \leq 4$ at $x = 0.352$ (or 0.351).

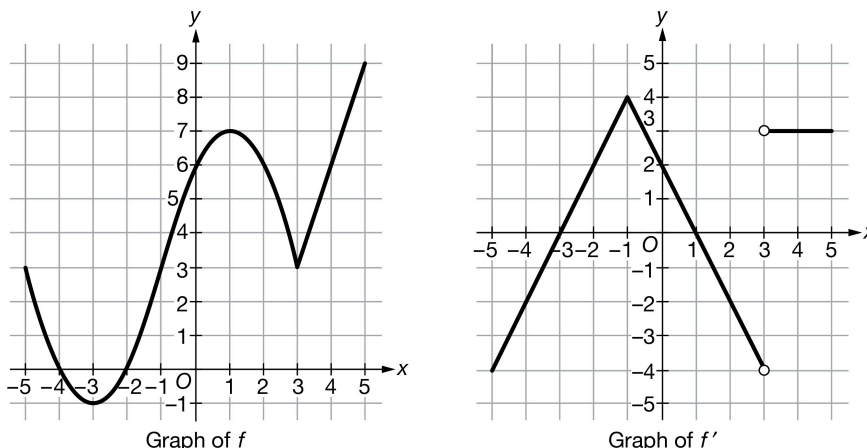
2-1-1

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the function f and its derivative f' are shown above for $-5 \leq x \leq 5$.

Find the average rate of change of f over the interval $-5 \leq x \leq 5$. For how many values of x in the interval $-5 \leq x \leq 5$ does the instantaneous rate of change of f equal the average rate of change of f over that interval?

Write an equation for the line tangent to the graph of f at $x = 2$.

For each of $\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)}$ and $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$, find the value or give a reason why it does not exist.

Let g be the function defined by $g(x) = f(x) \cdot \ln x$. Find $g'(4)$.

Part A

If the average rate of change has a maximum of one sign error, the response is eligible for the second point based on a correct interpretation of the graph of f' with the presented average rate of change.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

2-1-1

$$\frac{f(5)-f(-5)}{5-(-5)} = \frac{9-3}{5-(-5)} = \frac{6}{10} = \frac{3}{5}$$

The average rate of change of f over the interval $-5 \leq x \leq 5$ is $\frac{3}{5}$.

$f'(x)$, the instantaneous rate of change of f at x , equals $\frac{3}{5}$ for two values of x in the interval $-5 \leq x \leq 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct tangent line equation.

Solution:

$$f(2) = 6$$

$$f'(2) = -2$$

An equation of the line tangent to the graph of f at $x = 2$ is $y = 6 - 2(x - 2)$.

Part C

The second point is earned for “does not exist” with a reason that indicates f is not differentiable at that point (e.g., indicates graph has a corner point or sharp point).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow -1} \frac{f(x)-f(-1)}{x-(-1)}$
- ☐ $\lim_{x \rightarrow 3} \frac{f(x)-f(3)}{x-3}$ does not exist

Solution:

$$\lim_{x \rightarrow -1} \frac{f(x)-f(-1)}{x-(-1)} = f'(-1) = 4$$

$$\lim_{x \rightarrow 3} \frac{f(x)-f(3)}{x-3} \text{ does not exist because } f \text{ is not differentiable at } x = 3.$$

Part D

2-1-1

The first point is earned for a correct application of the product rule. The second point is earned for a correct expression that substitutes both correct numerical values.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ answer

Solution:

$$g'(x) = f'(x) \cdot \ln x + f(x) \cdot \frac{1}{x}$$

$$g'(4) = f'(4) \cdot \ln 4 + f(4) \cdot \frac{1}{4} = 3 \ln 4 + 6 \cdot \frac{1}{4} = 3 \ln 4 + \frac{3}{2}$$

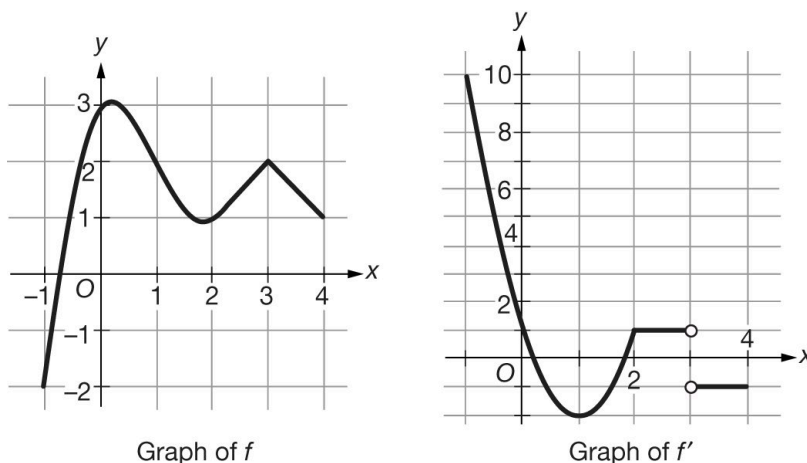
2-1-1

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the function f and its derivative f' are shown above for $-1 \leq x \leq 4$.

- Find the average rate of change of f over the interval $-1 \leq x \leq 4$. For how many values of x in the interval $-1 \leq x \leq 4$ does the instantaneous rate of change of f equal the average rate of change of f over that interval?
- Write an equation for the line tangent to the graph of f at $x = 1$.
- For each of $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$ and $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$, find the value or give a reason why it does not exist.
- Let g be the function defined by $g(x) = e^x f(x)$. Find $g'(0)$.

Part A

If the average rate of change has a maximum of one sign error, the response is eligible for the second point based on a correct interpretation of the graph of f' with the presented average rate of change.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

2-1-1

$$\frac{f(4)-f(-1)}{4-(-1)} = \frac{1-(-2)}{5} = \frac{3}{5}$$

The average rate of change of f over the interval is $\frac{3}{5}$

$f'(x)$, the instantaneous rate of change of f at x , equals $\frac{3}{5}$ for two values of x in the interval $[-1, 4]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct tangent line equation.

Solution:

$$f(1) = 2$$

$$f'(1) = -2$$

An equation of the line tangent to the graph of f at $x = 1$ is $y = 2 - 2(x - 1)$.

Part C

The second point is earned for “does not exist” with a reason that indicates f is not differentiable at that point (e.g., indicates graph has a corner point or sharp point).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow 2} \frac{f(x)-f(2)}{x-2}$
- ☐ $\lim_{x \rightarrow 3} \frac{f(x)-f(3)}{x-3}$ does not exist

Solution:

$$\lim_{x \rightarrow 2} \frac{f(x)-f(2)}{x-2} = f'(2) = 1$$

$$\lim_{x \rightarrow 3} \frac{f(x)-f(3)}{x-3} \text{ does not exist because } f \text{ is not differentiable at } x = 3.$$

Part D

2-1-1

The first point is earned for a correct application of the product rule. The second point is earned for a correct expression that substitutes both correct numerical values.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ answer

Solution:

$$g'(x) = e^x f(x) + e^x f'(x)$$

$$g'(0) = e^0 f(0) + e^0 f'(0) = 3 + 1 = 4$$

2-1-1

13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	2	2.1	2.2	2.3	2.4
$g'(x)$	16	25	36	49	64

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(2) = 5$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 2$ to approximate $g(2.2)$.
- Find the average rate of change of g' over the interval $2 \leq x \leq 2.4$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = (g(x))^3$. Find $h'(2)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_2^{2.4} g'(x) \, dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) > 0$ for $2 \leq x \leq 2.4$. Is the approximation from part (d) an overestimate or an underestimate for $\int_2^{2.4} g'(x) \, dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 2$ with two steps of equal size, to approximate $g(2.2)$. Show the computations that lead to your answer.
- It is known that $g''(2) = 80$ and $g'''(2) = 200$. Write the third-degree Taylor polynomial for g about $x = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $5 + 16 \cdot 0.2$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

2-1-1

$$g(2.2) \approx g(2) + g'(2)(2.2 - 2) = 5 + 16 \cdot 0.2 = 8.2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{64-16}{0.4}$ or $\frac{48}{2.4-2}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(2.4) - g'(2)}{2.4 - 2} = \frac{64 - 16}{0.4} = 120$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $3 \cdot 5^2 \cdot 16$.

The first point is for the correct application of the chain rule to produce the derivative $h'(x) = 3(g(x))^2 \cdot g'(x)$.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $3 \cdot 5^2 \cdot 16$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Chain rule

☐ Answer

Solution:

$$h'(x) = 3 \cdot (g(x))^2 \cdot g'(x)$$

$$h'(2) = 3 \cdot (g(2))^2 \cdot g'(2) = 3 \cdot 5^2 \cdot 16 = 1200$$

2-1-1

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $25 \cdot 0.2 + 49 \cdot 0.2$ or $(25 + 49) \cdot 0.2$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned}\int_2^{2.4} g'(x) \, dx &\approx g'(2.1)(2.2 - 2) + g'(2.3)(2.4 - 2.2) \\ &= 25 \cdot 0.2 + 49 \cdot 0.2 = 14.8\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave up given that $g'''(x)$ is positive.

The answer point is earned for using that $g'(x)$ is concave up on the given interval to conclude that the midpoint Riemann sum approximation of $\int_2^{2.4} g'(x) \, dx$ computed in part (d) must necessarily be an underestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an overestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

“Since $g'''(x) > 0$, then $g''(x)$ is increasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

“Since $g'''(x) > 0$, the tangent line to the graph of $g'(x)$ lies below $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

2-1-1

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

“Since $g'(x)$ is increasing and $g'''(x) > 0$, is greater than 0, $g'(x)$ is concave up.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.

“ $g'(x)$ is increasing. Since $g'''(x) > 0$, then $g'(x)$ is concave up.” This does earn the first point, because “ $g'(x)$ ” is not part of the necessary conditions.

“Since $g'''(x) > 0$, $g'(x)$ is concave up. Since it is also increasing, the approximation is an underestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

“The approximation is an underestimate because $g'(x)$ is increasing and concave up.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.



0	1	2
---	---	---

The student response accurately includes **one** of the criteria below.

- ☐ Concavity
- ☐ Answer

Solution:

Because $g'''(x) > 0$ for $2 \leq x \leq 2.4$, g' is concave up on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_2^{2.4} g'(x) dx$ is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler’s method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $5 + 16 \cdot 0.1 = 6.6$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 9.1 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $5 + 16 \cdot 0.1 + 25 \cdot 0.1$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
2	5	16	0.1	1.6	6.6
2.1	6.6	25	0.1	2.5	9.1

2-1-1

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(2.1) \approx g(2) + g'(2)(2.1 - 2) = 5 + 16 \cdot 0.1 = 6.6$$

$$g(2.2) \approx g(2.1) + g'(2.1)(2.2 - 2.1) \approx 6.6 + 25 \cdot 0.1 = 9.1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 2$.

Because $g''(2) = 80$ and $g'''(2) = 200$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(2)$ and $g'''(2)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(2) = 5$ given in the initial statement of the problem and $g'(2) = 16$ from the table for the coefficients involving those values. For example:

The expression $g(2) + g'(2)(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns only one point.

The expression $5 + 16(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names ($t, x, y, \dots$) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 2$ will earn no points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

2-1-1**Solution:**

$$\begin{aligned} &g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 \\ &= 5 + 16(x-2) + \frac{80}{2}(x-2)^2 + \frac{200}{6}(x-2)^3 \\ &= 5 + 16(x-2) + 40(x-2)^2 + \frac{100}{3}(x-2)^3 \end{aligned}$$

2-1-1

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	3	3.2	3.4	3.6	3.8
$g'(x)$	10	20	34	52	74

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(3) = 7$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 3$ to approximate $g(3.4)$.
- Find the average rate of change of g' over the interval $3 \leq x \leq 3.8$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = 4(g(x))^2$. Find $h'(3)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_3^{3.8} g'(x) dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) > 0$ for $3 \leq x \leq 3.8$. Is the approximation from part (d) an overestimate or an underestimate for $\int_3^{3.8} g'(x) dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 3$ with two steps of equal size, to approximate $g(3.4)$. Show the computations that lead to your answer.
- It is known that $g''(3) = 40$ and $g'''(3) = 100$. Write the third-degree Taylor polynomial for g about $x = 3$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $7 + 10 \cdot 0.4$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.



0

1

The student response accurately includes the criteria below.



Answer

Solution:

2-1-1

$$g(3.4) \approx g(3) + g'(3)(3.4 - 3) = 7 + 10 \cdot 0.4 = 11$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{74-10}{0.8}$ or $\frac{64}{3.8-3}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(3.8) - g'(3)}{3.8 - 3} = \frac{74 - 10}{0.8} = 80$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $8 \cdot 7 \cdot 10$.

The first point is for the correct application of the chain rule to produce the derivative $h'(x) = c \cdot 2 \cdot g(x) \cdot g'(x)$, where c is a non-zero constant. For example:

$\cdot 2 \cdot g(x) \cdot g'(x)$ earns the first point

$\cdot 8 \cdot g(x) \cdot g'(x)$ earns the first point.

$\cdot 3 \cdot g(x) \cdot g'(x)$ does not earn the first point since it is not of the correct form.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $8 \cdot 7 \cdot 10$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Chain rule

☐ Answer

2-1-1

Solution:

$$h'(x) = 8 \cdot g(x) \cdot g'(x)$$

$$h'(3) = 8 \cdot g(3) \cdot g'(3) = 8 \cdot 7 \cdot 10 = 560$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $20 \cdot 0.4 + 52 \cdot 0.4$ or $(20 + 52) \cdot 0.4$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned} \int_3^{3.8} g'(x) dx &\approx g'(3.2)(3.4 - 3) + g'(3.6)(3.8 - 3.4) \\ &= 20 \cdot 0.4 + 52 \cdot 0.4 = 28.8 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave up given that $g'''(x)$ is positive.

The answer point is earned for using that $g'(x)$ is concave up on the given interval to conclude that the midpoint Riemann sum approximation of $\int_3^{3.8} g'(x) dx$ computed in part (d) must necessarily be an underestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an overestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

2-1-1

“Since $g'''(x) > 0$, then $g''(x)$ is increasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

“Since $g'''(x) > 0$, the tangent line to the graph of $g'(x)$ lies below $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

“Since $g'(x)$ is increasing and $g'''(x) > 0$, $g'(x)$ is concave up.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.

“ $g'(x)$ is increasing. Since $g'''(x) > 0$, is greater than 0, then $g'(x)$ is concave up.” This does earn the first point, because “ $g'(x)$ is increasing” is not part of the necessary conditions.

“Since $g'''(x) > 0$, is greater than 0, $g'(x)$ is concave up. Since it is also increasing, the approximation is an underestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

“The approximation is an underestimate because $g'(x)$ is increasing and concave up.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Concavity
- ☐ Answer

Solution:

Because $g'''(x) > 0$ for $3 \leq x \leq 3.8$, g' is concave up on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_3^{3.8} g'(x) dx$ is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler’s method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $7 + 10 \cdot 0.2 = 9$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 13 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $7 + 10 \cdot 0.2 + 20 \cdot 0.2$ would earn both points.

Tabular responses:

2-1-1

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
3	7	10	0.2	2	9
3.2	9	20	0.2	4	13

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(3.2) \approx g(3) + g'(3)(3.2 - 3) = 7 + 10 \cdot 0.2 = 9$$

$$g(3.4) \approx g(3.2) + g'(3.2)(3.4 - 3.2) \approx 9 + 20 \cdot 0.2 = 13$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 3$.

Because $g''(3) = 40$ and $g'''(3) = 100$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(3)$ and $g'''(3)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(3) = 7$ given in the initial statement of the problem and $g'(3) = 10$ from the table for the coefficients involving those values. For example:

The expression $g(3) + g'(3)(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns only one point.

The expression $7 + 10(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names (t , x , y , ...) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 3$ will earn no points.

2-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} g(3) + g'(3)(x-3) + \frac{g''(3)}{2!}(x-3)^2 + \frac{g'''(3)}{3!}(x-3)^3 \\ = 7 + 10(x-3) + \frac{40}{2}(x-3)^2 + \frac{100}{6}(x-3)^3 \\ = 7 + 10(x-3) + 20(x-3)^2 + \frac{50}{3}(x-3)^3 \end{aligned}$$

2-1-1

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	4	4.2	4.4	4.6	4.8
$g'(x)$	70	48	30	16	6

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(4) = 5$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 4$ to approximate $g(4.8)$.
- Find the average rate of change of g' over the interval $4 \leq x \leq 4.8$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = 1 + (g(x))^2$. Find $h'(4)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_4^{4.8} g'(x) dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) > 0$ for $4 \leq x \leq 4.8$. Is the approximation from part (d) an overestimate or an underestimate for $\int_4^{4.8} g'(x) dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 4$ with two steps of equal size, to approximate $g(4.8)$. Show the computations that lead to your answer.
- It is known that $g''(4) = -120$ and $g'''(4) = 100$. Write the third-degree Taylor polynomial for g about $x = 4$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $5 + 70 \cdot 0.8$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

2-1-1

$$g(4.8) \approx g(4) + g'(4)(4.8 - 4) = 5 + 70 \cdot 0.8 = 61$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{6-70}{0.8}$ or $\frac{-64}{4.8-4}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(4.8) - g'(4)}{4.8 - 4} = \frac{6 - 70}{0.8} = -80$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $2 \cdot 5 \cdot 70$.

The first point is for the correct application of the chain rule on the term $(g(x))^2$ to produce the derivative $2 \cdot g(x) \cdot g'(x)$. If there are any errors in differentiating the constant term, the student does not earn the second point.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $2 \cdot 5 \cdot 70$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Chain rule

☐ Answer

Solution:

$$h'(x) = 2 \cdot g(x) \cdot g'(x)$$

$$h'(4) = 2 \cdot g(4) \cdot g'(4) = 2 \cdot 5 \cdot 70 = 700$$

2-1-1

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $48 \cdot 0.4 + 16 \cdot 0.4$ or $(48 + 16) \cdot 0.4$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned} \int_4^{4.8} g'(x) \, dx &\approx g'(4.2)(4.4 - 4) + g'(4.6)(4.8 - 4.4) \\ &= 48 \cdot 0.4 + 16 \cdot 0.4 = 25.6 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave up given that $g'''(x)$ is positive.

The answer point is earned for using that $g'(x)$ is concave up on the given interval to conclude that the midpoint Riemann sum approximation of $\int_4^{4.8} g'(x) \, dx$ computed in part (d) must necessarily be an underestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an overestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

· “Since $g'''(x) > 0$, then $g''(x)$ is increasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

· “Since $g'''(x) > 0$, the tangent line to the graph of $g'(x)$ lies below $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

2-1-1

Responses often conclude from the table of values that $g'(x)$ is decreasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

· “Since $g'(x)$ is decreasing and $g'''(x) > 0$, $g'(x)$ is concave up.” This does not earn the first point, because it implies that “ $g'(x)$ is decreasing” is part of the condition on the concavity of $g'(x)$.

· “ $g'(x)$ is decreasing. Since $g'''(x) > 0$, then $g'(x)$ is concave up.” This does earn the first point, because “ $g'(x)$ is decreasing” is not part of the necessary conditions.

· “Since $g'''(x) > 0$, $g'(x)$ is concave up. Since it is also decreasing, the approximation is an underestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is decreasing” is part of the rationale for the answer point.

· “The approximation is an underestimate because $g'(x)$ is decreasing and concave up.” This earns only the first point, because it implies that “ $g'(x)$ is decreasing” is part of the rationale for the answer point.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Concavity

☐ Answer

Solution:

Because $g'''(x) > 0$ for $4 \leq x \leq 4.8$, g' is concave up on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_4^{4.8} g'(x) dx$ is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler’s method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $5 + 70 \cdot 0.4 = 33$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 45 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $5 + 70 \cdot 0.4 + 30 \cdot 0.4$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

2-1-1

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
4	5	70	0.4	28	33
4.4	33	30	0.4	12	45

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(4.4) \approx g(4) + g'(4)(4.4 - 4) = 5 + 70 \cdot 0.4 = 33$$

$$g(4.8) \approx g(4.4) + g'(4.4)(4.8 - 4.4) \approx 33 + 30 \cdot 0.4 = 45$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 4$.

Because $g''(4) = -120$ and $g'''(4) = 100$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(4)$ and $g'''(4)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(4) = 5$ given in the initial statement of the problem and $g'(4) = 70$ from the table for the coefficients involving those values. For example:

· The expression $g(4) + g'(4)(x - 4) + \frac{g''(4)}{2!}(x - 4)^2 + \frac{g'''(4)}{3!}(x - 4)^3$ earns only one point.

· The expression $5 + 70(x - 4) + \frac{g''(4)}{2!}(x - 4)^2 + \frac{g'''(4)}{3!}(x - 4)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names (t , x , y , ...) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 4$ will earn no points.



0	1	2
---	---	---

2-1-1

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} g(4) + g'(4)(x-4) + \frac{g''(4)}{2!}(x-4)^2 + \frac{g'''(4)}{3!}(x-4)^3 \\ = 5 + 70(x-4) + \frac{-120}{2}(x-4)^2 + \frac{100}{6}(x-4)^3 \\ = 5 + 70(x-4) - 60(x-4)^2 + \frac{50}{3}(x-4)^3 \end{aligned}$$

2-1-1

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	3	3.1	3.2	3.3	3.4
$g'(x)$	14	25	34	41	46

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(3) = 10$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 3$ to approximate $g(3.4)$.
- Find the average rate of change of g' over the interval $3 \leq x \leq 3.4$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = (g(x))^3 - 1$. Find $h'(3)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_3^{3.4} g'(x) \, dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) < 0$ for $3 \leq x \leq 3.4$. Is the approximation from part (d) an overestimate or an underestimate for $\int_3^{3.4} g'(x) \, dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 3$ with two steps of equal size, to approximate $g(3.4)$. Show the computations that lead to your answer.
- It is known that $g''(3) = 120$ and $g'''(3) = -200$. Write the third-degree Taylor polynomial for g about $x = 3$.

Part A

10 + 14 · 0.4. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is 10 + 14 · 0.4.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

2-1-1

$$g(3.4) \approx g(3) + g'(3)(3.4 - 3) = 10 + 14 \cdot 0.4 = 15.6$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{46-14}{0.4}$ or $\frac{32}{3.4-3}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(3.4) - g'(3)}{3.4 - 3} = \frac{46 - 14}{0.4} = 80$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $3 \cdot 10^2 \cdot 14$.

The first point is for the correct application of the chain rule on the term $(g(x))^3$ to produce the derivative $3 \cdot (g(x))^2 \cdot g'(x)$. If there are any errors in differentiating the constant term, the student does not earn the second point.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $3 \cdot 10^2 \cdot 14$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Chain rule

☐ Answer

Solution:

$$h'(x) = 3 \cdot (g(x))^2 \cdot g'(x)$$

2-1-1

$$h'(3) = 3 \cdot (g(3))^2 \cdot g'(3) = 3 \cdot 10^2 \cdot 14 = 4200$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $25 \cdot 0.2 + 41 \cdot 0.2$ or $(25 + 41) \cdot 0.2$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned} \int_3^{3.4} g'(x) dx &\approx g'(3.1)(3.2 - 3) + g'(3.3)(3.4 - 3.2) \\ &= 25 \cdot 0.2 + 41 \cdot 0.2 = 13.2 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave down given that $g'''(x)$ is negative.

The answer point is earned for using that $g'(x)$ is concave down on the given interval to conclude that the midpoint Riemann sum approximation of $\int_3^{3.4} g'(x) dx$ computed in part (d) must necessarily be an overestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an underestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

· “Since $g'''(x) < 0$, then $g''(x)$ is decreasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

2-1-1

· “Since $g'''(x) < 0$, the tangent line to the graph of $g'(x)$ lies above $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

· “Since $g'(x)$ is increasing and $g'''(x) < 0$, $g'(x)$ is concave down.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.

· “ $g'(x)$ is increasing. Since $g'''(x) < 0$, then $g'(x)$ is concave down.” This does earn the first point, because “ $g'(x)$ is increasing” is not part of the necessary conditions.

· “Since $g'''(x) < 0$, $g'(x)$ is concave down. Since it is also increasing, the approximation is an overestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

· “The approximation is an overestimate because $g'(x)$ is increasing and concave down.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Concavity
- ☐ Answer

Solution:

Because $g'''(x) < 0$ for $3 \leq x \leq 3.4$, g' is concave down on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_3^{3.4} g'(x) dx$ is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler’s method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $10 + 14 \cdot 0.2 = 12.8$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 19.6 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $10 + 14 \cdot 0.2 + 34 \cdot 0.2$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

2-1-1

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
3	10	14	0.2	2.8	12.8
3.2	12.8	34	0.2	6.8	19.6

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(3.2) \approx g(3) + g'(3)(3.2 - 3) = 10 + 14 \cdot 0.2 = 12.8$$

$$g(3.4) \approx g(3.2) + g'(3.2)(3.4 - 3.2) \approx 12.8 + 34 \cdot 0.2 = 19.6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 3$.

Because $g''(3) = 120$ and $g'''(3) = -200$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(3)$ and $g'''(3)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(3) = 10$ given in the initial statement of the problem and $g'(3) = 14$ from the table for the coefficients involving those values. For example:

· The expression $g(3) + g'(3)(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns only one point.

· The expression $10 + 14(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names (t , x , y , ...) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “ $+ \dots$ ” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 3$ will earn no points.



0	1	2
---	---	---

2-1-1

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} & g(3) + g'(3)(x-3) + \frac{g''(3)}{2!}(x-3)^2 + \frac{g'''(3)}{3!}(x-3)^3 \\ &= 10 + 14(x-3) + \frac{120}{2}(x-3)^2 + \frac{-200}{6}(x-3)^3 \\ &= 10 + 14(x-3) + 60(x-3)^2 - \frac{100}{3}(x-3)^3 \end{aligned}$$

2-1-1

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	2	2.1	2.2	2.3	2.4
$g'(x)$	20	39	56	71	84

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(2) = 9$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 2$ to approximate $g(2.4)$.
- Find the average rate of change of g' over the interval $2 \leq x \leq 2.4$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = 6 + (g(x))^2$. Find $h'(2)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_2^{2.4} g'(x) dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) < 0$ for $2 \leq x \leq 2.4$. Is the approximation from part (d) an overestimate or an underestimate for $\int_2^{2.4} g'(x) dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 2$ with two steps of equal size, to approximate $g(2.4)$. Show the computations that lead to your answer.
- It is known that $g''(2) = 200$ and $g'''(2) = -200$. Write the third-degree Taylor polynomial for g about $x = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $9 + 20 \cdot 0.4$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

2-1-1

$$g(2.4) \approx g(2) + g'(2)(2.4 - 2) = 9 + 20 \cdot 0.4 = 17$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{84-20}{0.4}$ or $\frac{64}{2.4-2}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(2.4) - g'(2)}{2.4 - 2} = \frac{84 - 20}{0.4} = 160$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $2 \cdot 9 \cdot 20$.

The first point is for the correct application of the chain rule to produce the derivative $h'(x) = 2 \cdot g(x) \cdot g'(x)$.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $2 \cdot 9 \cdot 20$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Chain rule

☐ Answer

Solution:

$$h'(x) = 2 \cdot g(x) \cdot g'(x)$$

$$h'(2) = 2 \cdot g(2) \cdot g'(2) = 2 \cdot 9 \cdot 20 = 360$$

2-1-1

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $39 \cdot 0.2 + 71 \cdot 0.2$ or $(39 + 71) \cdot 0.2$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned}\int_2^{2.4} g'(x) dx &\approx g'(2.1)(2.2 - 2) + g'(2.3)(2.4 - 2.2) \\ &= 39 \cdot 0.2 + 71 \cdot 0.2 = 22\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave down given that $g'''(x)$ is negative.

The answer point is earned for using that $g'(x)$ is concave down on the given interval to conclude that the midpoint Riemann sum approximation of $\int_2^{2.4} g'(x) dx$ computed in part (d) must necessarily be an overestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an underestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

· “Since $g'''(x) < 0$, then $g''(x)$ is decreasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

· “Since $g'''(x) < 0$, the tangent line to the graph of $g'(x)$ lies above $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

2-1-1

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

· “Since $g'(x)$ is increasing and $g'''(x) < 0$, $g'(x)$ is concave down.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.

· “ $g'(x)$ is increasing. Since $g'''(x) < 0$, then $g'(x)$ is concave down.” This does earn the first point, because “ $g'(x)$ is increasing” is not part of the necessary conditions.

· “Since $g'''(x) < 0$, $g'(x)$ is concave down. Since it is also increasing, the approximation is an overestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

· “The approximation is an overestimate because $g'(x)$ is increasing and concave down.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Concavity

☐ Answer

Solution:

Because $g'''(x) < 0$ for $2 \leq x \leq 2.4$, g' is concave down on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_2^{2.4} g'(x) dx$ is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler’s method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $9 + 20 \cdot 0.2 = 13$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 24.2 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $9 + 20 \cdot 0.2 + 56 \cdot 0.2$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

2-1-1

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
2	9	20	0.2	4	13
2.2	13	56	0.2	11.2	24.2

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(2.2) \approx g(2) + g'(2)(2.2 - 2) = 9 + 20 \cdot 0.2 = 13$$

$$g(2.4) \approx g(2.2) + g'(2.2)(2.4 - 2.2) \approx 13 + 56 \cdot 0.2 = 24.2$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 2$.

Because $g''(2) = 200$ and $g'''(2) = -200$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(2)$ and $g'''(2)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(2) = 9$ given in the initial statement of the problem and $g'(2) = 20$ from the table for the coefficients involving those values. For example:

· The expression $g(2) + g'(2)(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns only one point.

· The expression $9 + 20(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names (t , x , y , ...) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 2$ will earn no points.



0	1	2
---	---	---

2-1-1

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 \\ = 9 + 20(x-2) + \frac{200}{2}(x-2)^2 + \frac{-200}{6}(x-2)^3 \\ = 9 + 20(x-2) + 100(x-2)^2 - \frac{100}{3}(x-2)^3 \end{aligned}$$

2-1-1

18. Let f be a function that is differentiable throughout its domain and that has the following properties.

(i) $f(x + y) = \frac{f(x) + f(y)}{1 - f(x)f(y)}$ for all real numbers x , y , and $x + y$ in the domain of f

(ii) $\lim_{h \rightarrow 0} f(h) = 0$

(iii) $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 1$

(a) Show that $f(0) = 0$.

(b) Use the definition of the derivative to show that $f'(x) = 1 + [f(x)]^2$. Indicate clearly where properties (i), (ii) and (iii) are used.

(c) Find $f(x)$ by solving the differential equation in part (b).

Part A

1 point is earned for to show $f(0) = 0$

Method 1:

f is continuous at 0, so $f(0) = \lim_{x \rightarrow 0} f(x) = 0$

Method 2:

$$f(0) = f(0 + 0) = \frac{f(0) + f(0)}{1 - f(0)f(0)}$$

$$f(0) (1 - [f(0)]^2) = 2f(0)$$

$$f(0) (-1 - [f(0)]^2) = 0$$

$$f(0) = 0$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for to show $f(0) = 0$

Method 1:

f is continuous at 0, so $f(0) = \lim_{x \rightarrow 0} f(x) = 0$

Method 2:

2-1-1

$$f(0) = f(0 + 0) = \frac{f(0)+f(0)}{1-f(0)f(0)}$$

$$f(0) \left(1 - [f(0)]^2\right) = 2f(0)$$

$$f(0) \left(-1 - [f(0)]^2\right) = 0$$

$$f(0) = 0$$

Part B

1 point is earned for definition of $f'(x)$

1 point is earned for clearly uses (i)

1 point is earned for clearly uses (ii)

1 point is earned for clearly uses (iii)

1 point is earned for algebraic manipulation to answer

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{f(x)+f(h)}{1-f(x)f(h)} - f(x)}{h} \quad [\text{By (i)}] \\ &= \lim_{h \rightarrow 0} \left[\frac{f(h)}{h} \cdot \frac{1+[f(x)]^2}{1-f(x)f(h)} \right] \\ &= 1 \cdot \frac{1+[f(x)]^2}{1-f(x) \cdot 0} \quad [\text{By (iii) \& (ii)}] \\ &= 1 + [f(x)]^2 \end{aligned}$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for definition of $f'(x)$

1 point is earned for clearly uses (i)

1 point is earned for clearly uses (ii)

1 point is earned for clearly uses (iii)

1 point is earned for algebraic manipulation to answer

2-1-1

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{f(x)+f(h)}{1-f(x)f(h)} - f(x)}{h} \quad [\text{By (i)}] \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(h)}{h} \cdot \frac{1+[f(x)]^2}{1-f(x)f(h)} \right] \\
 &= 1 \cdot \frac{1+[f(x)]^2}{1-f(x) \cdot 0} \quad [\text{By (iii) \& (ii)}] \\
 &= 1 + [f(x)]^2
 \end{aligned}$$

Part C

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

$$\text{Let } y = f(x); \frac{dy}{dx} = 1 + y^2$$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \quad [\text{or } C = n\pi, n \in \mathbb{Z}]$$

$$f(x) = \tan x \quad [\text{or } f(x) = \tan(x + n\pi)]$$

or

Method 2:

Guess that $f(x) = \tan x$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$

$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if separates variables

2-1-1

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

$$\text{Let } y = f(x); \frac{dy}{dx} = 1 + y^2$$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \text{ [or } C = n\pi, n \in \mathbb{Z}]$$

$$f(x) = \tan x \text{ [or } f(x) = \tan(x + n\pi)]$$

or

Method 2:

Guess that $f(x) = \tan x$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$

$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.

2-1-1

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	$0 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$	4	$4 < x < 6$	6
$g(x)$	1	Positive	4	Positive	6	Positive	8	Positive	4
$g'(x)$	3	Positive	0	Positive	4	Positive	0	Negative	-4
$g''(x)$	-4	Negative	0	Positive	0	Negative	-2	Negative	-2

Let g be a twice-differentiable function. The function g and its derivatives have the properties indicated in the table above.

(a) Find the average rate of change of g' over the interval $[0, 6]$. Show the computations that lead to your answer.

(b) Find the x -coordinate of each point of inflection of the graph of g on the open interval $0 < x < 6$. Justify your answers.

(c) Use a right Riemann sum with the three subintervals $[0, 2]$, $[2, 3]$, and $[3, 4]$ to approximate $\int_0^4 g(x) \, dx$. Show the computations that lead to your answer.

(d) Is the approximation from part (c) an overestimate or underestimate for $\int_0^4 g(x) \, dx$? Give a reason for your answer.

(e) Find the value of $\int_0^3 x \cdot g''(x) \, dx$. Show the computations that lead to your answer.

(f) Write the second-degree Taylor polynomial for g about $x = 4$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A correct numerical answer without supporting work does not earn this point.

The point is earned for a correct numerical difference in a correct quotient. The difference in the numerator must be shown either as $g'(6) - g'(0)$ or as $-4 - 3$.

The values of $g'(6)$ and $g'(0)$ must be pulled from the table; $\frac{g'(6) - g'(0)}{6}$ does not yet earn the point.

The answer does not need to be simplified but if simplified, it must be correct to earn the point.



0

1

The student response accurately includes the criteria below.

2-1-1

☐ Answer
Solution:

$$\text{Average rate of change} = \frac{g'(6) - g'(0)}{6 - 0} = \frac{-4 - 3}{6 - 0} = -\frac{7}{6}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for stating that g has points of inflection at both $x = 2$ and $x = 3$. No reasoning is needed to earn this point. Note that:

- Endpoints or points outside the interval $0 < x < 6$ do not affect the scoring. Responses are not penalized for inclusion and discussion of points outside of the interval.
- If any other internal points are identified as points of inflection or if only one point of inflection is given, the first point is not earned.
- If $g(2)$ and/or $g(3)$ are given as the points of inflection, the response does not earn the first point but is eligible for the second point.
- If ordered pairs are presented, they must be correct to earn the first point. That is, the ordered pairs must be $(2, g(2))$ and $(3, g(3))$, or equivalently $(2, 4)$ and $(3, 6)$.
- Set notation may be used: $\{2, 3\}$ should be read as “the set containing $x = 2$ and $x = 3$.” Using any other notation such as $[2, 3]$ does not earn the first point.
- Just writing $2, 3$ does not earn the point. The response must indicate that these are x -values.

Second point:

The second point is earned for the justification of both $x = 3$ and $x = 4$. The justification must state that $g''(x)$ changes sign at these places to earn the point.

The justification does not have to specify the direction of the sign change at either point. However, if the response reports the wrong direction of the sign change in $g''(x)$ at either point, this point is not earned.

Special case: If a response presents one of the two correct points of inflection (and presents no other points) with a correct justification, then the response will earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

2-1-1

$g''(x)$ changes sign at $x = 2$ and $x = 3$. Therefore, the graph of g has points of inflection at $x = 2$ and $x = 3$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for the setup of a right Riemann sum. The minimum requirement is that we must see the sum of a product of a height and a width for each of the three rectangles. The width of 1 can be implied and does not need to be shown.

The first point is earned if there is at most one error in the six factors of the terms of the sum
 $g(2)(2 - 0) + g(3)(3 - 2) + g(4)(4 - 3)$.

Responses that start with numerical values are eligible to earn the first point. For example, the following responses earn the first point:

$$\cdot 4 \cdot 2 + 6 \cdot 2 + 8 \cdot 1$$

$$\cdot 4 \cdot 2 + 6 + 8$$

$$\cdot a \cdot 2 + 6 + 8, \text{ where } a \neq 4$$

$$\cdot 4 \cdot b + 6 + 8, \text{ where } b \neq 2$$

In contrast, a response that only shows $8 + 6 + 8$ or 22 does not earn the first point.

Second point:

Only responses that earn the first point are eligible to earn the second point.

The second point is earned only for the correct answer (22). The answer does not need to be simplified, but if simplified, it must be correct to earn the second point.

Equating the actual value of the integral with the approximation will not result in a loss of a point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Right Riemann sum

☐ Answer

Solution:

$$\int_0^4 g(x) \, dx \approx g(2)(2 - 0) + g(3)(3 - 2) + g(4)(4 - 3)$$

$$= 4 \cdot 2 + 6 \cdot 1 + 8 \cdot 1 = 22$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1

Note: Responses in part (d) must include a conclusion with reasoning to be eligible for points. A correct conclusion without reasoning earns no points.

First point:

The first point is earned for reasoning that g is increasing over the given interval.

To earn the first point the response must state $g'(x) > 0$ or $g'(x) \geq 0$. For clarification:

- Stating g is increasing, without stating $g'(x) > 0$ or $g'(x) \geq 0$, does not earn the first point.
- Stating $g'(x) > 0$ or $g'(x) \geq 0$, without stating that g is increasing, earns the first point.

A picture, graph, or sign chart must be accompanied by a verbal explanation that $g'(x) > 0$ or $g'(x) \geq 0$ in order to be relevant. Without a verbal explanation, a picture, graph, or sign chart does not earn either point.

Use of pronouns such as “it” or the “curve” are ambiguous and will not earn the point unless they are clearly tied to a correct use of a word such as derivative, slope of g , g' , or g . Note that the word “function” will be read as meaning the function g .

Intervals are not required to earn the first point. If intervals are presented, they must be $0 < x < 2$ and $2 < x < 4$, or $0 < x < 4$, with or without endpoints included.

Second point:

The second point is earned for reasoning and a correct conclusion.

Eligibility: The response must state either that $g'(x) > 0$ or that g is increasing.

Special case: If a response includes $g'(x) > 0$, reports that g was decreasing, and concludes that the right Riemann sum approximation is an underestimate, the response earns the second point, but not the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ g increasing
- ☐ Answer

Solution:

$g'(x) > 0$ for $0 < x < 2$ and $2 < x < 4$, so g is increasing for $0 \leq x \leq 4$.

Therefore, the right Riemann sum approximation for $\int_0^4 g(x) dx$ is an overestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The first point is earned for using integration by parts and presenting an expression in x . Limits of integration are not considered for the first point. Responses with correct, incorrect, or no limits are eligible to earn the first point. A single sign error between the two

2-1-1

terms will still earn this point, but will not earn the third (answer) point. Responses that would earn the first point include:

$$\cdot x g'(x) \Big|_0^3 - \int_0^3 g'(x) dx$$

$$\cdot x g'(x) \Big|_0^3 + \int_0^3 g'(x) dx$$

$$\cdot x g'(x) - \int g'(x) dx$$

$$\cdot x g'(x) + \int g'(x) dx$$

$$\cdot x g'(x) \pm g(x)$$

Any mixture of indefinite and definite integrals in these expressions will also earn the first point.

Second point:

Responses are eligible for the second point even if they did not earn the first point. The second point is earned for correctly using the Fundamental Theorem of Calculus to conclude that $\int_0^3 g'(x) dx = g(3) - g(0)$ or $\int_0^3 g'(x) dx = 6 - 1$.

Note that it is not required to have the remaining terms from the integration by parts present or correct to earn this point.

A response that begins with a correct presentation of all terms resulting from evaluation of the integral by integration by parts will earn the integration by parts point and the Fundamental Theorem of Calculus point. Note that the term $0 \cdot g'(0)$ must be present.

Responses that begin with:

$$\cdot 3 \cdot g'(3) - 0 \cdot g'(0) - (g(3) - g(0)) \text{ earns the first two points.}$$

$$\cdot 3 \cdot g'(3) - 0 - (g(3) - g(0)) \text{ earns the first two points.}$$

$$\cdot 3 \cdot g'(3) - (g(3) - g(0)) \text{ does not earn the first point, but does earn the second point.}$$

$$\cdot 3 \cdot g'(3) - 0 \cdot g'(0) - (g(3) + g(0)) \text{ earns the first point, but does not earn the second point.}$$

Third point:

The third point is earned only for the correct answer (7). Simplifying the arithmetic is not needed but if presented, must be correct to earn the third point.

Responses that include a “+ C” on a definite integral are eligible for the first and second points (integration by parts and FTC) but cannot earn the answer point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integration by parts
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

2-1-1

Solution:

$$u = x \quad dv = g''(x) \, dx$$

$$du = dx \quad v = g'(x)$$

$$\int_0^3 x \cdot g''(x) \, dx = x \cdot g'(x) \Big|_0^3 - \int_0^3 g'(x) \, dx$$

$$= 3 \cdot g'(3) - 0 \cdot g'(0) - (g(3) - g(0))$$

$$= 3 \cdot 4 - 0 \cdot 3 - (6 - 1) = 12 - 0 - 5 = 7$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer. No work needs to be shown. Simplifications of the coefficients if presented, must be correct to earn the point.

A response with additional terms beyond the quadratic term or an ellipsis (+ $\cdots$) at the end of the expression will not earn this point.

Equating $g(x)$ with the polynomial is acceptable for this point.

Responses that present $8 - (x - 4)^2$ and then go on to expand the polynomial to $8 - 16 + 8x + x^2$ or $-8 + 8x + x^2$ will earn the point unless the expanded form is declared as the final answer by circling or boxing.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(4) + g'(4)(x - 4) + \frac{g''(4)}{2!}(x - 4)^2$$

$$= 8 + 0(x - 4) + \frac{-2}{2}(x - 4)^2 = 8 - (x - 4)^2$$

2-1-1

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	$0 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$	4	$4 < x < 6$	6
$g(x)$	5	Positive	1	Positive	3	Positive	5	Positive	8
$g'(x)$	-4	Negative	0	Positive	4	Positive	0	Positive	3
$g''(x)$	2	Positive	2	Positive	0	Negative	0	Positive	4

Let g be a twice-differentiable function. The function g and its derivatives have the properties indicated in the table above.

(a) Find the average rate of change of g' over the interval $[0, 6]$. Show the computations that lead to your answer.

(b) Find the x -coordinate of each point of inflection of the graph of g on the open interval $0 < x < 6$. Justify your answers.

(c) Use a left Riemann sum with the three subintervals $[2, 3]$, $[3, 4]$, and $[4, 6]$ to approximate $\int_2^6 g(x) \, dx$. Show the computations that lead to your answer.

(d) Is the approximation from part (c) an overestimate or underestimate for $\int_2^6 g(x) \, dx$? Give a reason for your answer.

(e) Find the value of $\int_0^3 x \cdot g''(x) \, dx$. Show the computations that lead to your answer.

(f) Write the second-degree Taylor polynomial for g about $x = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A correct numerical answer without supporting work does not earn this point.

The point is earned for a correct numerical difference in a correct quotient. The difference in the numerator must be shown either as $g'(6) - g'(0)$ or as $3 - (-4)$.

The values of $g'(6)$ and $g'(0)$ must be pulled from the table; $\frac{g'(6) - g'(0)}{6}$ does not yet earn the point.

The answer does not need to be simplified but if simplified, it must be correct to earn the point.



0

1

The student response accurately includes the criteria below.

2-1-1

☐ Answer
Solution:

$$\text{Average rate of change} = \frac{g'(6) - g'(0)}{6 - 0} = \frac{3 - (-4)}{6 - 0} = \frac{7}{6}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for stating that g has points of inflection at both $x = 3$ and $x = 4$. No reasoning is needed to earn this point. Note that:

- Endpoints or points outside the interval $0 < x < 6$ do not affect the scoring. Responses are not penalized for inclusion and discussion of points outside of the interval.
- If any other internal points are identified as points of inflection or if only one point of inflection is given, the first point is not earned.
- If $g(3)$ and/or $g(4)$ are given as the points of inflection, the response does not earn the first point but is eligible for the second point.
- If ordered pairs are presented, they must be correct to earn the first point. That is, the ordered pairs must be $(3, g(3))$ and $(4, g(4))$, or equivalently $(3, 3)$ and $(4, 5)$.
- Set notation may be used: $\{3, 4\}$ should be read as “the set containing $x = 3$ and $x = 4$.” Using any other notation such as $[3, 4]$ does not earn the first point.
- Just writing $3, 4$ does not earn the point. The response must indicate that these are x -values.

Second point:

The second point is earned for the justification of both $x = 3$ and $x = 4$. The justification must state that $g''(x)$ changes sign at these places to earn the point.

The justification does not have to specify the direction of the sign change at either point. However, if the response reports the wrong direction of the sign change in $g''(x)$ at either point, this point is not earned.

Special case: If a response presents one of the two correct points of inflection (and presents no other points) with a correct justification, then the response will earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

2-1-1

$g''(x)$ changes sign at $x = 3$ and $x = 4$. Therefore, the graph of g has points of inflection at $x = 3$ and $x = 4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for the setup of a left Riemann sum. The minimum requirement is that we must see the sum of a product of a height and a width for each of the three rectangles. The width of 1 can be implied and does not need to be shown.

The first point is earned if there is at most one error in the six factors of the terms of the sum
 $g(2)(3-2) + g(3)(4-3) + g(4)(6-4)$.

Responses that start with numerical values are eligible to earn the first point. For example, the following responses earn the first point:

$$\cdot 1 \cdot 2 + 3 \cdot 1 + 5 \cdot 2$$

$$\cdot 1 + 3 + 5 \cdot 2$$

$$\cdot 1 + 3 + a \cdot 2, \text{ where } a \neq 5$$

$$\cdot 1 + 3 + 5 \cdot b, \text{ where } b \neq 2$$

In contrast, a response that only shows $1 + 3 + 10$ or 14 does not earn the first point.

Second point:

Only responses that earn the first point are eligible to earn the second point.

The second point is earned only for the correct answer (14). The answer does not need to be simplified, but if simplified, it must be correct to earn the second point.

Equating the actual value of the integral with the approximation will not result in a loss of a point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Left Riemann sum

☐ Answer

Solution:

$$\int_2^6 g(x) \, dx \approx g(2)(3-2) + g(3)(4-3) + g(4)(6-4)$$

$$= 1 \cdot 1 + 3 \cdot 1 + 5 \cdot 2 = 14$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1

Note: Responses in part (d) must include a conclusion with reasoning to be eligible for points. A correct conclusion without reasoning earns no points.

First point:

The first point is earned for reasoning that g is increasing over the given interval.

To earn the first point the response must state $g'(x) > 0$ or $g'(x) \geq 0$. For clarification:

- Stating g is increasing, without stating $g'(x) > 0$ or $g'(x) \geq 0$, does not earn the first point.
- Stating $g'(x) > 0$ or $g'(x) \geq 0$, without stating that g is increasing, earns the first point.

A picture, graph, or sign chart must be accompanied by a verbal explanation that $g'(x) > 0$ or $g'(x) \geq 0$ in order to be relevant. Without a verbal explanation, a picture, graph, or sign chart does not earn either point.

Use of pronouns such as “it” or the “curve” are ambiguous and will not earn the point unless they are clearly tied to a correct use of a word such as derivative, slope of g , g' , or g . Note that the word “function” will be read as meaning the function g .

Intervals are not required to earn the first point. If intervals are presented, they must be $2 < x < 4$ and $4 < x < 6$, or $2 < x < 6$, with or without endpoints included.

Second point:

The second point is earned for reasoning and a correct conclusion.

Eligibility: The response must state either that $g'(x) > 0$ or that g is increasing.

Special case: If a response includes $g'(x) > 0$, reports that g was decreasing, and concludes that the left Riemann sum approximation is an overestimate, the response earns the second point, but not the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ g increasing
- ☐ Answer

Solution:

$g'(x) > 0$ for $2 < x < 4$ and $4 < x < 6$, so g is increasing for $2 \leq x \leq 6$.

Therefore, the left Riemann sum approximation for $\int_2^6 g(x) dx$ is an underestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The first point is earned for using integration by parts and presenting an expression in x . Limits of integration are not considered for the first point. Responses with correct, incorrect, or no limits are eligible to earn the first point. A single sign error between the two

2-1-1

terms will still earn this point, but will not earn the third (answer) point. Responses that would earn the first point include:

$$\cdot x g'(x) \Big|_0^3 - \int_0^3 g'(x) dx$$

$$\cdot x g'(x) \Big|_0^3 + \int_0^3 g'(x) dx$$

$$\cdot x g'(x) - \int g'(x) dx$$

$$\cdot x g'(x) + \int g'(x) dx$$

$$\cdot x g'(x) \pm g(x)$$

Any mixture of indefinite and definite integrals in these expressions will also earn the first point.

Second point:

Responses are eligible for the second point even if they did not earn the first point. The second point is earned for correctly using the Fundamental Theorem of Calculus to conclude that $\int_0^3 g'(x) dx = g(3) - g(0)$ or $\int_0^3 g'(x) dx = 3 - 5$.

Note that it is not required to have the remaining terms from the integration by parts present or correct to earn this point.

A response that begins with a correct presentation of all terms resulting from evaluation of the integral by integration by parts will earn the integration by parts point and the Fundamental Theorem of Calculus point. Note that the term $0 \cdot g'(0)$ must be present.

Responses that begin with:

$$\cdot 3 \cdot g'(3) - 0 \cdot g'(0) - (g(3) - g(0)) \text{ earns the first two points.}$$

$$\cdot 3 \cdot g'(3) - 0 - (g(3) - g(0)) \text{ earns the first two points.}$$

$$\cdot 3 \cdot g'(3) - (g(3) - g(0)) \text{ does not earn the first point, but does earn the second point.}$$

$$\cdot 3 \cdot g'(3) - 0 \cdot g'(0) - (g(3) + g(0)) \text{ earns the first point, but does not earn the second point.}$$

Third point:

The third point is earned only for the correct answer (14). Simplifying the arithmetic is not needed but if presented, must be correct to earn the third point.

Responses that include a “+ C” on a definite integral are eligible for the first and second points (integration by parts and FTC) but cannot earn the answer point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integration by parts
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

2-1-1

Solution:

$$u = x \quad dv = g''(x) dx$$

$$du = dx \quad v = g'(x)$$

$$\int_0^3 x \cdot g''(x) dx = x \cdot g'(x) \Big|_0^3 - \int_0^3 g'(x) dx$$

$$= 3 \cdot g'(3) - 0 \cdot g'(0) - (g(3) - g(0))$$

$$= 3 \cdot 4 - 0 \cdot (-4) - (3 - 5) = 12 - 0 + 2 = 14$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer. No work needs to be shown. Simplifications of the coefficients if presented, must be correct to earn the point.

A response with additional terms beyond the quadratic term or an ellipsis (+ $\dots$) at the end of the expression will not earn this point.

Equating $g(x)$ with the polynomial is acceptable for this point.

Responses that present $1 + (x - 2)^2$ and then go on to expand the polynomial to $1 + 4 - 4x + x^2$ or $5 - 4x + x^2$ will earn the point unless the expanded form is declared as the final answer by circling or boxing.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2$$

$$= 1 + 0(x-2) + \frac{2}{2}(x-2)^2 = 1 + (x-2)^2$$

2-1-1

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	$0 < x < 1$	1	$1 < x < 4$	4	$4 < x < 7$	7	$7 < x < 8$	8
$g(x)$	7	Positive	9	Positive	5	Positive	2	Positive	0
$g'(x)$	4	Positive	0	Negative	-2	Negative	0	Negative	-5
$g''(x)$	-2	Negative	-2	Negative	0	Positive	0	Negative	-3

Let g be a twice-differentiable function. The function g and its derivatives have the properties indicated in the table above.

(a) Find the average rate of change of g' over the interval $[0, 8]$. Show the computations that lead to your answer.

(b) Find the x -coordinate of each point of inflection of the graph of g on the open interval $0 < x < 8$. Justify your answers.

(c) Use a right Riemann sum with the three subintervals $[1, 4]$, $[4, 7]$, and $[7, 8]$ to approximate $\int_1^8 g(x) \, dx$. Show the computations that lead to your answer.

(d) Is the approximation from part (c) an overestimate or underestimate for $\int_1^8 g(x) \, dx$? Give a reason for your answer.

(e) Find the value of $\int_0^4 x \cdot g''(x) \, dx$. Show the computations that lead to your answer.

(f) Write the second-degree Taylor polynomial for g about $x = 1$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A correct numerical answer without supporting work does not earn this point.

The point is earned for a correct numerical difference in a correct quotient. The difference in the numerator must be shown either as $g'(8) - g'(0)$ or as $-5 - 4$.

The values of $g'(8)$ and $g'(0)$ must be pulled from the table; $\frac{g'(8) - g'(0)}{8}$ does not yet earn the point.

The answer does not need to be simplified but if simplified, it must be correct to earn the point.



0

1

The student response accurately includes the criteria below.

2-1-1

☐ Answer
Solution:

$$\text{Average rate of change} = \frac{g'(8) - g'(0)}{8 - 0} = \frac{-5 - 4}{8 - 0} = -\frac{9}{8}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for stating that g has points of inflection at both $x = 4$ and $x = 7$. No reasoning is needed to earn this point. Note that:

- Endpoints or points outside the interval $0 < x < 8$ do not affect the scoring. Responses are not penalized for inclusion and discussion of points outside of the interval.
- If any other internal points are identified as points of inflection or if only one point of inflection is given, the first point is not earned.
- If $g(4)$ and/or $g(7)$ are given as the points of inflection, the response does not earn the first point but is eligible for the second point.
- If ordered pairs are presented, they must be correct to earn the first point. That is, the ordered pairs must be $(4, g(4))$ and $(7, g(7))$, or equivalently $(4, 5)$ and $(7, 2)$.
- Set notation may be used: $\{4, 7\}$ should be read as “the set containing $x = 4$ and $x = 7$. ” Using any other notation such as $[4, 7]$ does not earn the first point.
- Just writing 4, 7 does not earn the point. The response must indicate that these are x -values.

Second point:

The second point is earned for the justification of both $x = 4$ and $x = 7$. The justification must state that $g''(x)$ changes sign at these places to earn the point.

The justification does not have to specify the direction of the sign change at either point. However, if the response reports the wrong direction of the sign change in $g''(x)$ at either point, this point is not earned.

Special case: If a response presents one of the two correct points of inflection (and presents no other points) with a correct justification, then the response will earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

2-1-1

$g''(x)$ changes sign at $x = 4$ and $x = 7$. Therefore, the graph of g has points of inflection at $x = 4$ and $x = 7$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for the setup of a right Riemann sum. The minimum requirement is that we must see the sum of a product of a height and a width for each of the three rectangles, with a single exception noted below. The third rectangle has height 0; the area of the third rectangle must be expressed either as a product or as a 0 unless there are no errors in the two remaining terms.

The first point is earned if there is at most one error in the six factors of the terms of the sum

$$g(4)(4-1) + g(7)(7-4) + g(8)(8-7).$$

Responses that start with numerical values are eligible to earn the first point. For example, the following responses earn the first point:

$$\cdot 5 \cdot 2 + 2 \cdot 3 + 0 \cdot 1$$

$$\cdot 5 \cdot 3 + 2 \cdot 1 + 0$$

$$\cdot 5 \cdot 3 + 2 \cdot 3 + 0 \cdot a, \text{ where } a \neq 1$$

$$\cdot 5 \cdot 3 + 2 \cdot 3$$

In contrast, a response that only shows $15 + 6 + 0$ or 21 does not earn the first point.

Second point:

Only responses that earn the first point are eligible to earn the second point.

The second point is earned only for the correct answer (21). The answer does not need to be simplified, but if simplified, it must be correct to earn the second point.

Equating the actual value of the integral with the approximation will not result in a loss of a point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Right Riemann sum

☐ Answer

Solution:

$$\int_1^8 g(x) dx \approx g(4)(4-1) + g(7)(7-4) + g(8)(8-7)$$

$$= 5 \cdot 3 + 2 \cdot 3 + 0 \cdot 1 = 21$$

Part D

2-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: Responses in part (d) must include a conclusion with reasoning to be eligible for points. A correct conclusion without reasoning earns no points.

First point:

The first point is earned for reasoning that g is decreasing over the given interval.

To earn the first point the response must state $g'(x) < 0$ or $g'(x) \leq 0$. For clarification:

- Stating g is decreasing, without stating $g'(x) < 0$ or $g'(x) \leq 0$, does not earn the first point.
- Stating $g'(x) < 0$ or $g'(x) \leq 0$, without stating that g is decreasing, earns the first point.

A picture, graph, or sign chart must be accompanied by a verbal explanation that $g'(x) < 0$ or $g'(x) \leq 0$ in order to be relevant. Without a verbal explanation, a picture, graph, or sign chart does not earn either point.

Use of pronouns such as “it” or the “curve” are ambiguous and will not earn the point unless they are clearly tied to a correct use of a word such as derivative, slope of g , g' , or g . Note that the word “function” will be read as meaning the function g .

Intervals are not required to earn the first point. If intervals are presented, they must be $1 < x < 7$ and $7 < x < 8$, or $1 < x < 8$, with or without endpoints included.

Second point:

The second point is earned for reasoning and a correct conclusion.

Eligibility: The response must state either that $g'(x) < 0$ or that g is decreasing.

Special case: If a response includes $g'(x) < 0$, reports that g was increasing, and concludes that the right Riemann sum approximation is an overestimate, the response earns the second point, but not the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ g decreasing
- ☐ Answer

Solution:

$g'(x) < 0$ for $1 < x < 7$ and $7 < x < 8$, so g is decreasing for $1 \leq x \leq 8$.

Therefore, the right Riemann sum approximation for $\int_1^8 g(x) dx$ is an underestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

2-1-1

The first point is earned for using integration by parts and presenting an expression in x . Limits of integration are not considered for the first point. Responses with correct, incorrect, or no limits are eligible to earn the first point. A single sign error between the two terms will still earn this point, but will not earn the third (answer) point. Responses that would earn the first point include:

$$\cdot x g'(x) \Big|_0^4 - \int_0^4 g'(x) dx$$

$$\cdot x g'(x) \Big|_0^4 + \int_0^4 g'(x) dx$$

$$\cdot x g'(x) - \int g'(x) dx$$

$$\cdot x g'(x) + \int g'(x) dx$$

$$\cdot x g'(x) \pm g(x)$$

Any mixture of indefinite and definite integrals in these expressions will also earn the first point.

Second point:

Responses are eligible for the second point even if they did not earn the first point. The second point is earned for correctly using the Fundamental Theorem of Calculus to conclude that $\int_0^4 g'(x) dx = g(4) - g(0)$ or $\int_0^4 g'(x) dx = 5 - 7$.

Note that it is not required to have the remaining terms from the integration by parts present or correct to earn this point.

A response that begins with a correct presentation of all terms resulting from evaluation of the integral by integration by parts will earn the integration by parts point and the Fundamental Theorem of Calculus point. Note that the term $0 \cdot g'(0)$ must be present.

Responses that begin with:

$$\cdot 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) - g(0)) \text{ earns the first two points.}$$

$$\cdot 4 \cdot g'(4) - 0 - (g(4) - g(0)) \text{ earns the first two points.}$$

$$\cdot 4 \cdot g'(4) - (g(4) - g(0)) \text{ does not earn the first point, but does earn the second point.}$$

$$\cdot 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) + g(0)) \text{ earns the first point, but does not earn the second point.}$$

Third point:

The third point is earned only for the correct answer (-6). Simplifying the arithmetic is not needed but if presented, must be correct to earn the third point.

Responses that include a “ $+C$ ” on a definite integral are eligible for the first and second points (integration by parts and FTC) but cannot earn the answer point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

☐

Integration by parts

2-1-1

☐ Fundamental Theorem of Calculus

☐ Answer
Solution:

$$u = x \quad dv = g''(x) dx$$

$$du = dx \quad v = g'(x)$$

$$\int_0^4 x \cdot g''(x) dx = x \cdot g'(x) \Big|_0^4 - \int_0^4 g'(x) dx$$

$$= 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) - g(0))$$

$$= 4 \cdot (-2) - 0 \cdot 4 - (5 - 7) = -8 - 0 + 2 = -6$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer. No work needs to be shown. Simplifications of the coefficients if presented, must be correct to earn the point.

A response with additional terms beyond the quadratic term or an ellipsis (+ $\cdots$) at the end of the expression will not earn this point.

Equating $g(x)$ with the polynomial is acceptable for this point.

Responses that present $9 - (x - 1)^2$ and then go on to expand the polynomial to $9 - 1 + 2x - x^2$ or $8 + 2x - x^2$ will earn the point unless the expanded form is declared as the final answer by circling or boxing.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer
Solution:

$$g(1) + g'(1)(x-1) + \frac{g''(1)}{2!}(x-1)^2$$

$$= 9 + 0(x-1) + \frac{-2}{2}(x-1)^2 = 9 - (x-1)^2$$

2-1-1

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	$0 < x < 1$	1	$1 < x < 4$	4	$4 < x < 7$	7	$7 < x < 8$	8
$g(x)$	10	Positive	8	Positive	5	Positive	1	Positive	3
$g'(x)$	-5	Negative	0	Negative	-2	Negative	0	Positive	4
$g''(x)$	3	Positive	0	Negative	0	Positive	2	Positive	2

Let g be a twice-differentiable function. The function g and its derivatives have the properties indicated in the table above.

(a) Find the average rate of change of g' over the interval $[0, 8]$. Show the computations that lead to your answer.

(b) Find the x -coordinate of each point of inflection of the graph of g on the open interval $0 < x < 8$. Justify your answers.

(c) Use a left Riemann sum with the three subintervals $[0, 1]$, $[1, 4]$, and $[4, 7]$ to approximate $\int_0^7 g(x) \, dx$. Show the computations that lead to your answer.

(d) Is the approximation from part (c) an overestimate or underestimate for $\int_0^7 g(x) \, dx$? Give a reason for your answer.

(e) Find the value of $\int_0^4 x \cdot g''(x) \, dx$. Show the computations that lead to your answer.

(f) Write the second-degree Taylor polynomial for g about $x = 7$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A correct numerical answer without supporting work does not earn this point.

The point is earned for a correct numerical difference in a correct quotient. The difference in the numerator must be shown either as $g'(8) - g'(0)$ or as $4 - (-5)$.

The values of $g'(8)$ and $g'(0)$ must be pulled from the table; $\frac{g'(8) - g'(0)}{8}$ does not yet earn the point.

The answer does not need to be simplified but if simplified, it must be correct to earn the point.



0

1

The student response accurately includes the criteria below.

2-1-1

☐ Answer
Solution:

$$\text{Average rate of change} = \frac{g'(8) - g'(0)}{8 - 0} = \frac{4 - (-5)}{8 - 0} = \frac{9}{8}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for stating that g has points of inflection at both $x = 1$ and $x = 4$. No reasoning is needed to earn this point. Note that:

- Endpoints or points outside the interval $0 < x < 8$ do not affect the scoring. Responses are not penalized for inclusion and discussion of points outside of the interval.
- If any other internal points are identified as points of inflection or if only one point of inflection is given, the first point is not earned.
- If $g(1)$ and/or $g(4)$ are given as the points of inflection, the response does not earn the first point but is eligible for the second point.
- If ordered pairs are presented, they must be correct to earn the first point. That is, the ordered pairs must be $(1, g(1))$ and $(4, g(4))$, or equivalently $(1, 8)$ and $(4, 5)$.
- Set notation may be used: $\{1, 4\}$ should be read as “the set containing $x = 1$ and $x = 4$.” Using any other notation such as $[1, 4]$ does not earn the first point.
- Just writing 1, 4 does not earn the point. The response must indicate that these are x -values.

Second point:

The second point is earned for the justification of both $x = 1$ and $x = 4$. The justification must state that $g''(x)$ changes sign at these places to earn the point.

The justification does not have to specify the direction of the sign change at either point. However, if the response reports the wrong direction of the sign change in $g''(x)$ at either point, this point is not earned.

Special case: If a response presents one of the two correct points of inflection (and presents no other points) with a correct justification, then the response will earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

2-1-1

$g''(x)$ changes sign at $x = 1$ and $x = 4$. Therefore, the graph of g has points of inflection at $x = 1$ and $x = 4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for the setup of a left Riemann sum. The minimum requirement is that we must see the sum of a product of a height and a width for each of the three rectangles. The width of 1 can be implied and does not need to be shown.

The first point is earned if there is at most one error in the six factors of the terms of the sum
 $g(0)(1-0) + g(1)(4-1) + g(4)(7-4)$.

Responses that start with numerical values are eligible to earn the first point. For example, the following responses earn the first point:

$$\cdot 10 \cdot 1 + 8 \cdot 3 + 5 \cdot 3$$

$$\cdot 10 + 8 \cdot 3 + 5 \cdot 3$$

$$\cdot 10 + 8 \cdot 3 + a \cdot 3, \text{ where } a \neq 5$$

$$\cdot 10 + 8 \cdot 3 + 5 \cdot b, \text{ where } b \neq 3$$

In contrast, a response that only shows $10 + 24 + 15$ or 49 does not earn the first point.

Second point:

Only responses that earn the first point are eligible to earn the second point.

The second point is earned only for the correct answer (49). The answer does not need to be simplified, but if simplified, it must be correct to earn the second point.

Equating the actual value of the integral with the approximation will not result in a loss of a point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Left Riemann sum

☐ Answer

Solution:

$$\int_0^7 g(x) dx \approx g(0)(1-0) + g(1)(4-1) + g(4)(7-4)$$

$$= 10 \cdot 1 + 8 \cdot 3 + 5 \cdot 3 = 49$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1

Note: Responses in part (d) must include a conclusion with reasoning to be eligible for points. A correct conclusion without reasoning earns no points.

First point:

The first point is earned for reasoning that g is decreasing over the given interval.

To earn the first point the response must state $g'(x) < 0$ or $g'(x) \leq 0$. For clarification:

- Stating g is decreasing, without stating $g'(x) < 0$ or $g'(x) \leq 0$, does not earn the first point.
- Stating $g'(x) < 0$ or $g'(x) \leq 0$, without stating that g is decreasing, earns the first point.

A picture, graph, or sign chart must be accompanied by a verbal explanation that $g'(x) < 0$ or $g'(x) \leq 0$ in order to be relevant. Without a verbal explanation, a picture, graph, or sign chart does not earn either point.

Use of pronouns such as “it” or the “curve” are ambiguous and will not earn the point unless they are clearly tied to a correct use of a word such as derivative, slope of g , g' , or g . Note that the word “function” will be read as meaning the function g .

Intervals are not required to earn the first point. If intervals are presented, they must be $0 < x < 1$ and $1 < x < 7$, or $0 < x < 7$, with or without endpoints included.

Second point:

The second point is earned for reasoning and a correct conclusion.

Eligibility: The response must state either that $g'(x) < 0$ or that g is decreasing.

Special case: If a response includes $g'(x) < 0$, reports that g was increasing, and concludes that the left Riemann sum approximation is an underestimate, the response earns the second point, but not the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ g decreasing
- ☐ Answer

Solution:

$g'(x) < 0$ for $0 < x < 1$ and $1 < x < 7$, so g is decreasing for $0 \leq x \leq 7$.

Therefore, the left Riemann sum approximation for $\int_0^7 g(x) dx$ is an overestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The first point is earned for using integration by parts and presenting an expression in x . Limits of integration are not considered for the first point. Responses with correct, incorrect, or no limits are eligible to earn the first point. A single sign error between the two

2-1-1

terms will still earn this point, but will not earn the third (answer) point. Responses that would earn the first point include:

$$\cdot x g'(x) \Big|_0^4 - \int_0^4 g'(x) dx$$

$$\cdot x g'(x) \Big|_0^4 + \int_0^4 g'(x) dx$$

$$\cdot x g'(x) - \int g'(x) dx$$

$$\cdot x g'(x) + \int g'(x) dx$$

$$\cdot x g'(x) \pm g(x)$$

Any mixture of indefinite and definite integrals in these expressions will also earn the first point.

Second point:

Responses are eligible for the second point even if they did not earn the first point. The second point is earned for correctly using the Fundamental Theorem of Calculus to conclude that $\int_0^4 g'(x) dx = g(4) - g(0)$ or $\int_0^4 g'(x) dx = 5 - 10$.

Note that it is not required to have the remaining terms from the integration by parts present or correct to earn this point.

A response that begins with a correct presentation of all terms resulting from evaluation of the integral by integration by parts will earn the integration by parts point and the Fundamental Theorem of Calculus point. Note that the term $0 \cdot g'(0)$ must be present.

Responses that begin with:

$\cdot 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) - g(0))$ earns the first two points.

$\cdot 4 \cdot g'(4) - 0 - (g(4) - g(0))$ earns the first two points.

$\cdot 4 \cdot g'(4) - (g(4) - g(0))$ does not earn the first point, but does earn the second point.

$\cdot 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) + g(0))$ earns the first point, but does not earn the second point.

Third point:

The third point is earned only for the correct answer (-3). Simplifying the arithmetic is not needed but if presented, must be correct to earn the third point.

Responses that include a “ $+C$ ” on a definite integral are eligible for the first and second points (integration by parts and FTC) but cannot earn the answer point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Integration by parts
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

2-1-1

Solution:

$$u = x \quad dv = g''(x) \, dx$$

$$du = dx \quad v = g'(x)$$

$$\int_0^4 x \cdot g''(x) \, dx = x \cdot g'(x) \Big|_0^4 - \int_0^4 g'(x) \, dx$$

$$= 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) - g(0))$$

$$= 4 \cdot (-2) - 0 \cdot (-5) - (5 - 10) = -8 - 0 + 5 = -3$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer. No work needs to be shown. Simplifications of the coefficients if presented, must be correct to earn the point.

A response with additional terms beyond the quadratic term or an ellipsis (+ ···) at the end of the expression will not earn this point.

Equating $g(x)$ with the polynomial is acceptable for this point.

Responses that present $1 + (x - 7)^2$ and then go on to expand the polynomial to $1 + x^2 - 14x + 49$ or $x^2 - 14x + 50$ will earn the point unless the expanded form is declared as the final answer by circling or boxing.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(7) + g'(7)(x - 7) + \frac{g''(7)}{2!}(x - 7)^2$$

$$= 1 + 0(x - 7) + \frac{2}{2}(x - 7)^2 = 1 + (x - 7)^2$$

2-1-1

23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	$0 < x < 3$	3	$3 < x < 4$	4	$4 < x < 5$	5	$5 < x < 8$	8
$g(x)$	10	Positive	6	Positive	4	Positive	1	Positive	5
$g'(x)$	-3	Negative	0	Negative	-5	Negative	0	Positive	2
$g''(x)$	3	Positive	0	Negative	0	Positive	2	Positive	1

Let g be a twice-differentiable function. The function g and its derivatives have the properties indicated in the table above.

(a) Find the average rate of change of g' over the interval $[0, 8]$. Show the computations that lead to your answer.

(b) Find the x -coordinate of each point of inflection of the graph of g on the open interval $0 < x < 8$. Justify your answers.

(c) Use a right Riemann sum with the three subintervals $[0, 3]$, $[3, 4]$, and $[4, 5]$ to approximate $\int_0^5 g(x) \, dx$. Show the computations that lead to your answer.

(d) Is the approximation from part (c) an overestimate or underestimate for $\int_0^5 g(x) \, dx$? Give a reason for your answer.

(e) Find the value of $\int_0^4 x \cdot g''(x) \, dx$. Show the computations that lead to your answer.

(f) Write the second-degree Taylor polynomial for g about $x = 5$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A correct numerical answer without supporting work does not earn this point.

The point is earned for a correct numerical difference in a correct quotient. The difference in the numerator must be shown either as $g'(8) - g'(0)$ or as $2 - (-3)$.

The values of $g'(8)$ and $g'(0)$ must be pulled from the table; $\frac{g'(8) - g'(0)}{8}$ does not yet earn the point.

The answer does not need to be simplified but if simplified, it must be correct to earn the point.



0

1

The student response accurately includes the criteria below.

2-1-1

☐ Answer
Solution:

$$\text{Average rate of change} = \frac{g'(8) - g'(0)}{8 - 0} = \frac{2 - (-3)}{8 - 0} = \frac{5}{8}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for stating that g has points of inflection at both $x = 3$ and $x = 4$. No reasoning is needed to earn this point. Note that:

- Endpoints or points outside the interval $0 < x < 8$ do not affect the scoring. Responses are not penalized for inclusion and discussion of points outside of the interval.
- If any other internal points are identified as points of inflection or if only one point of inflection is given, the first point is not earned.
- If $g(3)$ and/or $g(4)$ are given as the points of inflection, the response does not earn the first point but is eligible for the second point.
- If ordered pairs are presented, they must be correct to earn the first point. That is, the ordered pairs must be $(3, g(3))$ and $(4, g(4))$, or equivalently $(3, 6)$ and $(4, 4)$.
- Set notation may be used: $\{3, 4\}$ should be read as “the set containing $x = 3$ and $x = 4$. ” Using any other notation such as $[3, 4]$ does not earn the first point.
- Just writing 3, 4 does not earn the point. The response must indicate that these are x -values.

Second point:

The second point is earned for the justification of both $x = 3$ and $x = 4$. The justification must state that $g''(x)$ changes sign at these places to earn the point.

The justification does not have to specify the direction of the sign change at either point. However, if the response reports the wrong direction of the sign change in $g''(x)$ at either point, this point is not earned.

Special case: If a response presents one of the two correct points of inflection (and presents no other points) with a correct justification, then the response will earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

2-1-1

$g''(x)$ changes sign at $x = 3$ and $x = 4$. Therefore, the graph of g has points of inflection at $x = 3$ and $x = 4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The first point is earned for the setup of a right Riemann sum. The minimum requirement is that we must see the sum of a product of a height and a width for each of the three rectangles. The width of 1 can be implied and does not need to be shown.

The first point is earned if there is at most one error in the six factors of the terms of the sum

$$g(3)(3-0) + g(4)(4-3) + g(5)(5-4).$$

Responses that start with numerical values are eligible to earn the first point. For example, the following responses earn the first point:

$$\cdot 6 \cdot 2 + 4 \cdot 1 + 1 \cdot 1$$

$$\cdot 6 \cdot 3 + 4 + 1$$

$$\cdot a \cdot 3 + 4 + 1, \text{ where } a \neq 6$$

$$\cdot 6 \cdot b + 4 + 1, \text{ where } b \neq 3$$

In contrast, a response that only shows $18 + 4 + 1$ or 23 does not earn the first point.

Second point:

Only responses that earn the first point are eligible to earn the second point.

The second point is earned only for the correct answer (23). The answer does not need to be simplified, but if simplified, it must be correct to earn the second point.

Equating the actual value of the integral with the approximation will not result in a loss of a point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Right Riemann sum

☐ Answer

Solution:

$$\int_0^5 g(x) dx \approx g(3)(3-0) + g(4)(4-3) + g(5)(5-4)$$

$$= 6 \cdot 3 + 4 \cdot 1 + 1 \cdot 1 = 23$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1

Note: Responses in part (d) must include a conclusion with reasoning to be eligible for points. A correct conclusion without reasoning earns no points.

First point:

The first point is earned for reasoning that g is decreasing over the given interval.

To earn the first point the response must state $g'(x) < 0$ or $g'(x) \leq 0$. For clarification:

- Stating g is decreasing, without stating $g'(x) < 0$ or $g'(x) \leq 0$, does not earn the first point.
- Stating $g'(x) < 0$ or $g'(x) \leq 0$, without stating that g is decreasing, earns the first point.

A picture, graph, or sign chart must be accompanied by a verbal explanation that $g'(x) < 0$ or $g'(x) \leq 0$ in order to be relevant. Without a verbal explanation, a picture, graph, or sign chart does not earn either point.

Use of pronouns such as “it” or the “curve” are ambiguous and will not earn the point unless they are clearly tied to a correct use of a word such as derivative, slope of g , g' , or g . Note that the word “function” will be read as meaning the function g .

Intervals are not required to earn the first point. If intervals are presented, they must be $0 < x < 3$ and $3 < x < 5$, or $0 < x < 5$, with or without endpoints included.

Second point:

The second point is earned for reasoning and a correct conclusion.

Eligibility: The response must state either that $g'(x) < 0$ or that g is decreasing.

Special case: If a response includes $g'(x) < 0$, reports that g was increasing, and concludes that the right Riemann sum approximation is an overestimate, the response earns the second point, but not the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ g decreasing
- ☐ Answer

Solution:

$g'(x) < 0$ for $0 < x < 3$ and $3 < x < 5$, so g is decreasing for $0 \leq x \leq 5$.

Therefore, the right Riemann sum approximation for $\int_0^5 g(x) dx$ is an underestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The first point is earned for using integration by parts and presenting an expression in x . Limits of integration are not considered for the first point. Responses with correct, incorrect, or no limits are eligible to earn the first point. A single sign error between the two

2-1-1

terms will still earn this point, but will not earn the third (answer) point. Responses that would earn the first point include:

$$\cdot x g'(x) \Big|_0^4 - \int_0^4 g'(x) dx$$

$$\cdot x g'(x) \Big|_0^4 + \int_0^4 g'(x) dx$$

$$\cdot x g'(x) - \int g'(x) dx$$

$$\cdot x g'(x) + \int g'(x) dx$$

$$\cdot x g'(x) \pm g(x)$$

Any mixture of indefinite and definite integrals in these expressions will also earn the first point.

Second point:

Responses are eligible for the second point even if they did not earn the first point. The second point is earned for correctly using the Fundamental Theorem of Calculus to conclude that $\int_0^4 g'(x) dx = g(4) - g(0)$ or $\int_0^4 g'(x) dx = 4 - 10$.

Note that it is not required to have the remaining terms from the integration by parts present or correct to earn this point.

A response that begins with a correct presentation of all terms resulting from evaluation of the integral by integration by parts will earn the integration by parts point and the Fundamental Theorem of Calculus point. Note that the term $0 \cdot g'(0)$ must be present.

Responses that begin with:

$$\cdot 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) - g(0)) \text{ earns the first two points.}$$

$$\cdot 4 \cdot g'(4) - 0 - (g(4) - g(0)) \text{ earns the first two points.}$$

$$\cdot 4 \cdot g'(4) - (g(4) - g(0)) \text{ does not earn the first point, but does earn the second point.}$$

$$\cdot 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) + g(0)) \text{ earns the first point, but does not earn the second point.}$$

Third point:

The third point is earned only for the correct answer (-14). Simplifying the arithmetic is not needed but if presented, must be correct to earn the third point.

Responses that include a “ $+C$ ” on a definite integral are eligible for the first and second points (integration by parts and FTC) but cannot earn the answer point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integration by parts
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

2-1-1

Solution:

$$u = x \quad dv = g''(x) \, dx$$

$$du = dx \quad v = g'(x)$$

$$\int_0^4 x \cdot g''(x) \, dx = x \cdot g'(x) \Big|_0^4 - \int_0^4 g'(x) \, dx$$

$$= 4 \cdot g'(4) - 0 \cdot g'(0) - (g(4) - g(0))$$

$$= 4 \cdot (-5) - 0 \cdot (-3) - (4 - 10) = -20 - 0 + 6 = -14$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer. No work needs to be shown. Simplifications of the coefficients if presented, must be correct to earn the point.

A response with additional terms beyond the quadratic term or an ellipsis (+ $\dots$) at the end of the expression will not earn this point.

Equating $g(x)$ with the polynomial is acceptable for this point.

Responses that present $1 + (x - 5)^2$ and then go on to expand the polynomial to $1 + 25 - 10x + x^2$ or $26 - 10x + x^2$ will earn the point unless the expanded form is declared as the final answer by circling or boxing.



0	1
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The student response accurately includes the criteria below.

☐

Answer

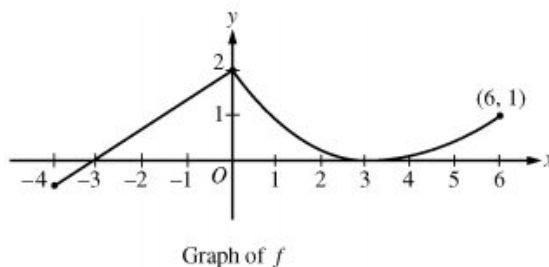
Solution:

$$g(5) + g'(5)(x - 5) + \frac{g''(5)}{2!}(x - 5)^2$$

$$= 1 + 0(x - 5) + \frac{2}{2}(x - 5)^2 = 1 + (x - 5)^2$$

2-1-1

24.



A continuous function f is defined on the closed interval $-4 \leq x \leq 6$. The graph of f consists of a line segment and a curve that is tangent to the x -axis at $x = 3$, as shown in the figure above. On the interval $0 < x < 6$, the function f is twice differentiable, with $f''(x) > 0$.

- (a) Is f differentiable at $x = 0$? Use the definition of the derivative with one-sided limits to justify your answer.
- (b) For how many values of a , $-4 \leq a < 6$, is the average rate of change of f on the interval $[a, 6]$ equal to 0? Give a reason for your answer.
- (c) Is there a value of a , $-4 \leq a < 6$, for which the Mean Value Theorem, applied to the interval $[a, 6]$, guarantees a value c , $a < c < 6$, at which $f'(c) = \frac{1}{3}$? Justify your answer.
- (d) The function g is defined by $g(x) = \int_0^x f(t) dt$ for $-4 \leq x \leq 6$. On what intervals contained in $[-4, 6]$ is the graph of g concave up? Explain your reasoning.

Part A

1 point is earned if sets up difference quotient at $x = 0$

1 point is earned for the answer with justification

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree, f is not differentiable at $x = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if sets up difference quotient at $x = 0$

1 point is earned for the answer with justification

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree, f is not differentiable at $x = 0$.

2-1-1

Part B

1 point is earned for expression of average rate of change

1 point is earned for answer with reason

$\frac{f(6)-f(a)}{6-a} = 0$ when $f(a) = f(6)$. There are two values of a for which this is true.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for expression of average rate of change

1 point is earned for answer with reason

$\frac{f(6)-f(a)}{6-a} = 0$ when $f(a) = f(6)$. There are two values of a for which this is true.

Part C

1 point is earned for answering "yes" and identifying $a = 3$

1 point is earned for the justification

Yes, $a = 3$. The function f is differentiable on the interval $3 < x < 6$ and continuous on $3 \leq x \leq 6$.

Also, $\frac{f(6)-f(3)}{6-3} = \frac{1-0}{6-3} = \frac{1}{3}$.

By the Mean Value Theorem, there is a value c , $3 < c < 6$, such that $f'(c) = \frac{1}{3}$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for answering "yes" and identifying $a = 3$

1 point is earned for the justification

Yes, $a = 3$. The function f is differentiable on the interval $3 < x < 6$ and continuous on $3 \leq x \leq 6$.

Also, $\frac{f(6)-f(3)}{6-3} = \frac{1-0}{6-3} = \frac{1}{3}$.

By the Mean Value Theorem, there is a value c , $3 < c < 6$, such that $f'(c) = \frac{1}{3}$.

Part D

1 point is earned for $g'(x) = f(x)$

2-1-1

1 point is earned for considering $g''(x) > 0$

1 point is earned for the answer

$$g'(x) = f(x), g''(x) = f'(x)$$

$$g''(x) > 0 \text{ when } f'(x) > 0$$

This is true for $-4 < x < 0$ and $3 < x < 6$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$

1 point is earned for considering $g''(x) > 0$

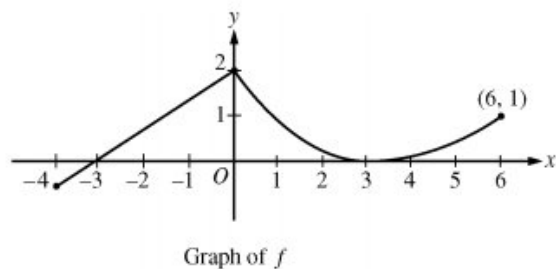
1 point is earned for the answer

$$g'(x) = f(x), g''(x) = f'(x)$$

$$g''(x) > 0 \text{ when } f'(x) > 0$$

This is true for $-4 < x < 0$ and $3 < x < 6$.

2-1-1



A continuous function f is defined on the closed interval $-4 \leq x \leq 6$. The graph of f consists of a line segment and a curve that is tangent to the x -axis at $x=3$, as shown in the figure above. On the interval $0 < x < 6$, the function f is twice differentiable, with $f''(x) > 0$.

25. Is f differentiable at $x=0$? Use the definition of the derivative with one-sided limits to justify your answer.

Part A

1 point is earned if sets up difference quotient at $x=0$

1 point is earned for the answer with justification

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree, f is not differentiable at $x=0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if sets up difference quotient at $x=0$

1 point is earned for the answer with justification

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree, f is not differentiable at $x=0$.

2-1-1

26. For how many values of a , $-4 \leq a < 6$, is the average rate of change of f on the interval $[a, 6]$ equal to 0? Give a reason for your answer.
-

Part B

1 point is earned for expression of average rate of change

1 point is earned for answer with reason

$$\frac{f(6)-f(a)}{6-a} = 0 \text{ when } f(a)=f(6). \text{ There are two values of } a \text{ for which this is true.}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for expression of average rate of change

1 point is earned for answer with reason

$$\frac{f(6)-f(a)}{6-a} = 0$$

There are two values of a for which this is true.

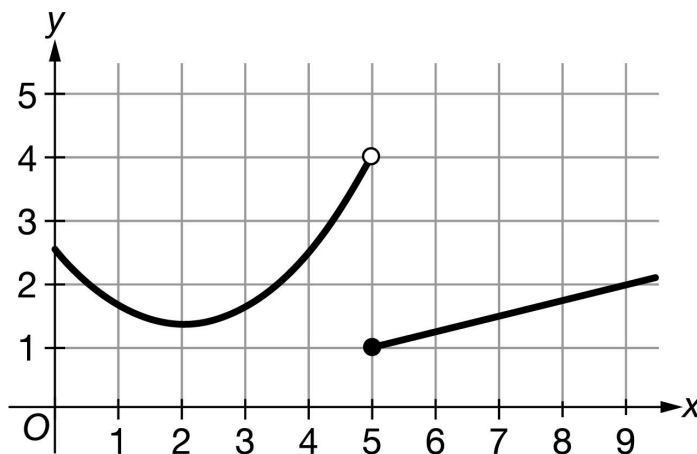
2-1-1

27. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

x	3	3.5	4	4.5	5	5.5	6
$f(x)$	1.653	2.02	2.533	3.193	1	1.125	1.25

The graph of the function f and a table of selected values of $f(x)$ are shown above. The graph of f has a horizontal tangent line at $x = 2$, is concave up for $0 < x < 5$, and is linear for $x \geq 5$.

- (a) Approximate the value of $f'(4.5)$ using data from the table. Show the computations that lead to your answer.
- (b) Is there a value of x , for $0 < x < 5$, such that $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = 0$? Give a reason for your answer.
- (c) For each of the following limits, find the value or explain why it does not exist.
- (i) $\lim_{h \rightarrow 0^+} \frac{f(5+h)-f(5)}{h}$
- (ii) $\lim_{h \rightarrow 0^-} \frac{f(5+h)-f(5)}{h}$
- (iii) $\lim_{h \rightarrow 0} \frac{f(5+h)-f(5)}{h}$

Part A

The first point is earned for a correct difference quotient over a subinterval of $3 \leq x < 5$ that contains $x = 4.5$. A response using $(5, 1)$ as the right endpoint earns only 1 point for knowing to use a difference quotient; this response is not eligible for the second point because $x = 5$ is not in the relevant portion of the domain. The second point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ difference quotient
- ☐ approximation

Solution:

$$f'(4.5) \approx \frac{f(4.5) - f(4)}{4.5 - 4} = \frac{3.193 - 2.533}{4.5 - 4} = \frac{0.660}{0.5} = 1.320$$

Part B

The first point is for recognizing that the given expression is the derivative of f . The second point is for the correct answer (there is a value of x , for $0 < x < 5$) and communicating the connection between the derivative and the slope of the tangent line at $x = 2$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$
- ☐ answer with reason

Solution:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

$f'(x) = 0$ at a point where the line tangent to the graph of f has slope 0. Since the graph of f has a horizontal tangent at $x = 2$, there is a value of x , for $0 < x < 5$, such that $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0$.

Part C

To earn the first point, the response must report the slope of the linear section of the graph. To earn the second point, the response must explain that the difference in the numerator of the expression will approach 3 as h approaches 0.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

2-1-1

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{h \rightarrow 0^+} \frac{f(5+h) - f(5)}{h}$
- ☐ $\lim_{h \rightarrow 0^-} \frac{f(5+h) - f(5)}{h}$ does not exist with explanation
- ☐ $\lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h}$ does not exist with explanation

Solution:

$$\lim_{h \rightarrow 0^+} \frac{f(5+h) - f(5)}{h} = \frac{1}{4}$$

$$\lim_{h \rightarrow 0^-} \frac{f(5+h) - f(5)}{h} \text{ does not exist because the numerator approaches 3 as the denominator approaches 0.}$$

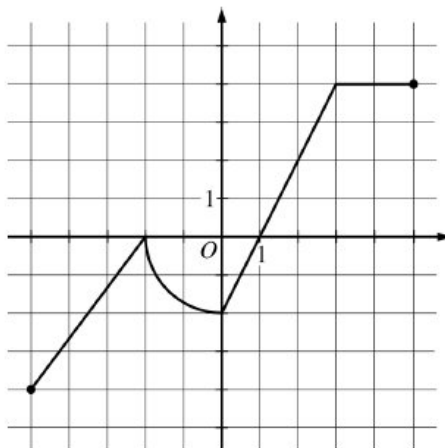
$$\lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} \text{ does not exist because the left-hand limit in (ii) does not exist.}$$

—OR—

Because f is not continuous at $x = 5$ due to a jump discontinuity, f is not differentiable at $x = 5$. Since $\lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h}$ is the definition of the derivative of f at $x = 5$, $\lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h}$ does not exist.

2-1-1

28. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of f

The graph of the function f , consisting of three line segments and a quarter of a circle, is shown above. Let g be the function defined by $g(x) = \int_1^x f(t) dt$.

- Find the average rate of change of g from $x = -5$ to $x = 5$.
- Find the instantaneous rate of change of g with respect to x at $x = 3$, or state that it does not exist.
- On what open intervals, if any, is the graph of g concave up? Justify your answer.
- Find all x -values in the interval $-5 < x < 5$ at which g has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

Part A

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$$

Part B

1 point is earned for: answer

2-1-1

$$g'(x) = f(x)$$

$$g'(3) = f(3) = 4$$

The instantaneous rate of change of g at $x = 3$ is 4.



0	1
---	---

The student response earns all of the following points:

1 point is earned for: answer

$$g'(x) = f(x)$$

$$g'(3) = f(3) = 4$$

The instantaneous rate of change of g at $x = 3$ is 4.

Part C

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.

Part D

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$$g'(x) = f(x) \text{ is defined at all } x \text{ with } -5 < x < 5.$$

$$g'(x) = f(x) = 0 \text{ at } x = -2 \text{ and } x = 1.$$

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

2-1-1



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

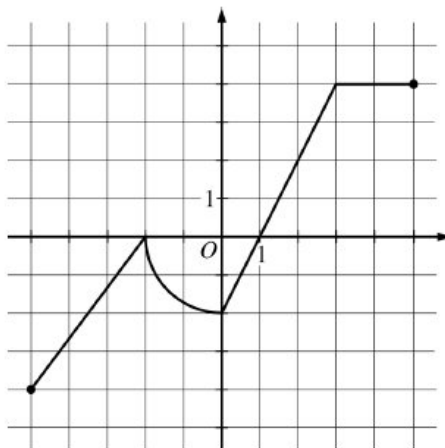
Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

2-1-1

29. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of f

The graph of the function f , consisting of three line segments and a quarter of a circle, is shown above. Let g be the function defined by $g(x) = \int_1^x f(t) dt$.

- Find the average rate of change of g from $x = -5$ to $x = 5$.
- Find the instantaneous rate of change of g with respect to x at $x = 3$, or state that it does not exist.
- On what open intervals, if any, is the graph of g concave up? Justify your answer.
- Find all x -values in the interval $-5 < x < 5$ at which g has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

Part A

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$$

Part B

1 point is earned for: answer

$$g'(x) = f(x)$$

2-1-1

$$g'(3) = f(3) = 4$$

The instantaneous rate of change of g at $x = 3$ is 4.



0	1
---	---

The student response earns all of the following points:

1 point is earned for: answer

$$g'(x) = f(x)$$

$$g'(3) = f(3) = 4$$

The instantaneous rate of change of g at $x = 3$ is 4.

Part C

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.

Part D

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

2-1-1



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

2-1-1

30. Let $F(x) = \int_0^x \sin(t^2) dt$ for $0 \leq x \leq 3$.

If the average rate of change of F on the closed interval $[1, 3]$ is k , find $\int_1^3 \sin(t^2) dt$ in terms of k .

Part C

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$

31. What is the average rate of change of $y = \cos(2x)$ on the interval $[0, \frac{\pi}{2}]$?

(A) $-\frac{4}{\pi}$

(B) -1

(C) 0

(D) $\frac{\sqrt{2}}{2}$

(E) $\frac{4}{\pi}$



32. What is the average rate of change of the function f given by $f(x) = x^4 - 5x$ on the closed interval $[0, 3]$?

(A) 8.5

(B) 8.7

(C) 22

(D) 33

(E) 66



2-1-1

33.

x	0	2	4	6	8	10
$f(x)$	5	7	8	0	-15	-20

Let f be a differentiable function with selected values given in the table above. What is the average rate of change of f over the closed interval $0 \leq x \leq 10$?

(A) -6

(B) $-\frac{5}{2}$

(C) -2

(D) $-\frac{2}{5}$ (E) $\frac{2}{5}$ 

2-1-1

34. Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by $F(t) = 82 + 4 \sin\left(\frac{t}{2}\right)$ for $0 \leq t \leq 30$, where $F(t)$ is measured in cars per minute and t is measured in minutes.



What is the average rate of change of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.

Part D

1 point is earned for the answer

$$\frac{F(15) - F(10)}{15 - 10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{F(15) - F(10)}{15 - 10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$

Units

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)



0	1
---	---

The student response earns all of the following points:

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)

35. If f is the function given by $f(x) = 3x^2 - x^3$, then the average rate of change of f on the closed interval $[1, 5]$ is

(A) -21

(B) -13

(C) -12

(D) -9



2-1-1

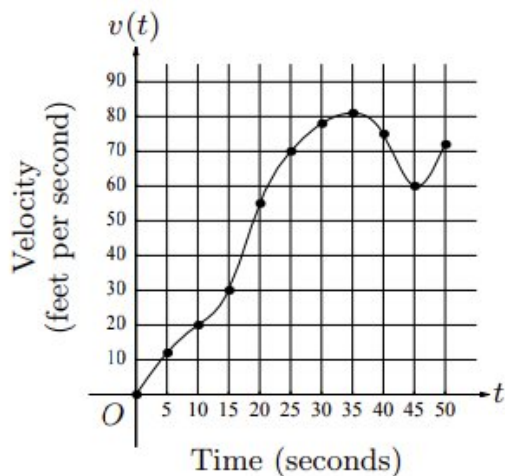
36.  The function f is given by $f(x) = \int_1^x \sqrt{t^3 + 2} dt$. What is the average rate of change of f over the interval $[0, 3]$?

- (A) 1.324
(B) 1.497
(C) 1.696
(D) 2.266
(E) 2.694



2-1-1

37.



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph.

Find the average acceleration of the car, in ft/sec^2 , over the interval $0 \leq t \leq 50$.

Part B

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2



0

1

The student response earns one of the following points:

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2

2-1-1

38. A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S , where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.



According to the model, what is the average rate of change in the height of water in the can over the 60-day period? Show the computations that lead to your answer. Indicate units of measure.

Part B

1 point is earned for the answer

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the answer

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$

2-1-1

39. A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 3t^2 - 2t - 1$. The position $x(t)$ is 5 for $t = 2$.

For what values of t , $0 \leq t \leq 3$, is the particle's instantaneous velocity the same as its average velocity on the closed interval $[0, 3]$?

Part B

1 point is earned for the correct average velocity = 5

$$\text{avg. vel.} = \frac{x(3) - x(0)}{3 - 0}$$

$$= \frac{18 - 3}{3} = 5$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 = \text{an avg. velocity in } [0, 3]$

$$t = \frac{1 + \sqrt{19}}{3} \text{ or } 1.786$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct average velocity = 5

$$\text{avg. vel.} = \frac{x(3) - x(0)}{3 - 0}$$

$$= \frac{18 - 3}{3} = 5$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 = \text{an avg. velocity in } [0, 3]$

$$t = \frac{1 + \sqrt{19}}{3} \text{ or } 1.786$$

2-1-1

40. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for answer with reason

$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because, $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for answer with reason

$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because, $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

2-1-1

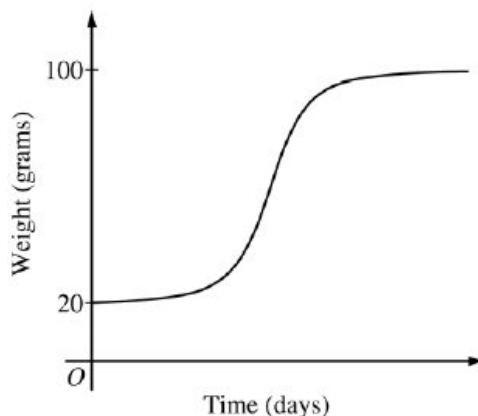
41. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

(a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

(b) Find $\frac{d^2B}{dt^2}$ in terms of B . Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



(c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

2-1-1

$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

Part B

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5} \frac{dB}{dt} = -\frac{1}{5} \cdot \frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5} \frac{dB}{dt} = -\frac{1}{5} \cdot \frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

Part C

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

2-1-1

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \frac{1}{5}t + c$$

$$\text{Because } 20 \leq B < 100, |100 - B| = 100 - B.$$

$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5}, t \geq 0$$



0	1	2	3	4	5
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The student earns all of the following points:

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \frac{1}{5}t + c$$

$$\text{Because } 20 \leq B < 100, |100 - B| = 100 - B.$$

$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5}, t \geq 0$$

2-1-1

42.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.



Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.

Part A

The response can earn up to 2 points:

1 point: For correct approximation

1 point: For correct units

$$a(1.4) \approx \frac{v(1.6) - v(1.2)}{1.6 - 1.2} = \frac{16.3 - 17.2}{1.6 - 1.2} = -2.25 \text{ miles/hr}^2$$



0	1	2
---	---	---

The response earns both of the following points:

1 point: For correct approximation

1 point: For correct units

$$a(1.4) \approx \frac{v(1.6) - v(1.2)}{1.6 - 1.2} = \frac{16.3 - 17.2}{1.6 - 1.2} = -2.25 \text{ miles/hr}^2$$

2-1-1

43.



t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$ hours, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.

(a) Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{2.4} v(t) dt$ in the context of the problem. Approximate $\int_0^{2.4} v(t) dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.

(c) For $0 \leq t \leq 2.4$ hours, Ruth's velocity can be modeled by the function g given by $g(t) = \frac{24t+5\sin(6t)}{t+0.7}$. According to the model, what was Ruth's average velocity during the time interval $0 \leq t \leq 2.4$?

(d) According to the model, $g(t) = \frac{24t+5\sin(6t)}{t+0.7}$, is Ruth's speed increasing or decreasing at time $t = 1.3$? Give a reason for your answer.

Part A

The response can earn up to 2 points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6)-v(1.2)}{1.6-1.2} = \frac{16.3-17.2}{1.6-1.2} = -2.25 \text{ miles/hr}^2$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6)-v(1.2)}{1.6-1.2} = \frac{16.3-17.2}{1.6-1.2} = -2.25 \text{ miles/hr}^2$$

Part B

The response can earn up to 3 points:

1 point is earned for the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point is earned for use of the midpoint Riemann sum

2-1-1

1 point is earned for the correct approximation

$$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) \\ = 36.64 \text{ miles}$$

✓

0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for the correct interpretation

$$\int_0^{2.4} v(t) dt \text{ is the total distance Ruth traveled, in miles, from time } t = 0 \text{ to time } t = 2.4 \text{ hours.}$$

1 point is earned for use of the midpoint Riemann sum

1 point is earned for the correct approximation

$$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) \\ = 36.64 \text{ miles}$$

Part C

The response can earn up to 2 points:

1 point is earned for the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point is earned for the correct answer

$$\text{Average velocity} = 14.064 \text{ miles/hr}$$

✓

0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point is earned for the correct answer

$$\text{Average velocity} = 14.064 \text{ miles/hr}$$

2-1-1**Part D**

The response can earn up to 2 points:

2 points are earned for correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.



0	1	2
---	---	---

The response earns all of the following points:

2 points are earned for correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.

2-1-1

44. The wind chill is the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v , in miles per hour (mph). If the air temperature is 32°F , then the wind chill is given by $W(v)=55.6-22.1v^{0.16}$ and is valid for $5 \leq v \leq 60$.



Find the average rate of change of W over the interval $5 \leq v \leq 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \leq v \leq 60$.

Part A

1 point is earned for the average rate of change.

1 point is earned for $W'(v) = \text{average rate of change}$

1 point is earned for the value of v .

The average rate of change of W over the interval

$$5 \leq v \leq 60 \text{ is } \frac{W(60)-W(5)}{60-5} = -0.253 \text{ or } -0.254.$$

$$W'(v) = \frac{W(60)-W(5)}{60-5} \text{ when } v=23.011.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the average rate of change.

1 point is earned for $W'(v) = \text{average rate of change}$

1 point is earned for the value of v .

The average rate of change of W over the interval

$$5 \leq v \leq 60 \text{ is } \frac{W(60)-W(5)}{60-5} = -0.253 \text{ or } -0.254.$$

$$W'(v) = \frac{W(60)-W(5)}{60-5} \text{ when } v=23.011.$$

45. For values of h very close to 0, which of the following functions best approximates $f(x) = \frac{\tan(x+h) - \tan x}{h}$?

(A) $\sin x$

(B) $\frac{\sin x}{x}$

(C) $\frac{\tan x}{x}$

(D) $\sec x$

(E) $\sec^2 x$



2-1-1

46. Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.



Find the average rate of change of $A(t)$ over the interval $0 \leq t \leq 30$. Indicate units of measure.

Part A

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$

47. For time $t \geq 1$, the position of a particle moving along the x-axis is given by $p(t) = \sqrt{t} - 2$. At what time t in the interval $1 \leq t \leq 16$ is the instantaneous velocity of the particle equal to the average velocity of the particle over the interval $1 \leq t \leq 16$

(A) 1

(B) $\frac{121}{25}$

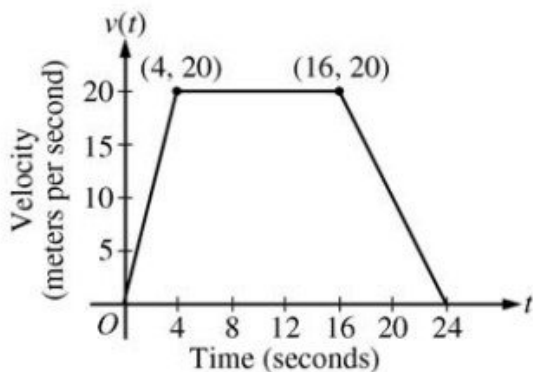
(C) $\frac{25}{4}$

(D) 25



2-1-1

48.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

Find the average rate of change of v over the interval $8 \leq t \leq 20$. Does the Mean Value Theorem guarantee a value of c , for $8 < c < 20$, such that $v'(c)$ is equal to this average rate of change? Why or why not?

Part D

1 point is earned for the average rate of change of v on $[8, 20]$

1 point is earned for the answer with explanation

The average rate of change of v on $[8, 20]$ is

$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} m/sec^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at $t = 16$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the average rate of change of v on $[8, 20]$

1 point is earned for the answer with explanation

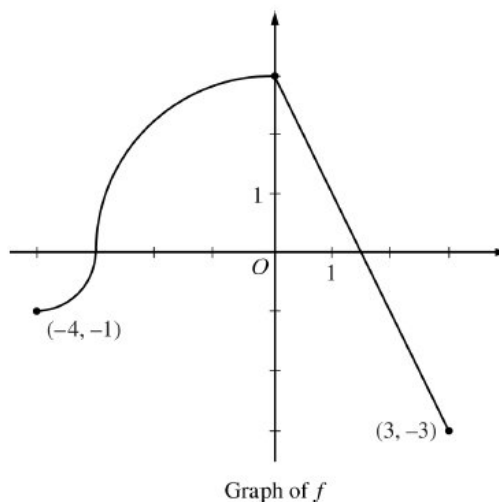
The average rate of change of v on $[8, 20]$ is

$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} m/sec^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at $t = 16$.

2-1-1

49.



The continuous function f is defined on the interval $-4 \leq x \leq 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t) dt$.

Find the average rate of change of f on the interval $-4 \leq x \leq 3$. There is no point c , $-4 < c < 3$, for which $f'(c)$ is equal to that average rate of change. Explain why this statement does not contradict the Mean Value Theorem.

Part D

1 point is earned for average rate of change

1 point is earned for explanation

The average rate of change of f on the interval

$$-4 \leq x \leq 3 \text{ is } \frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$$

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for average rate of change

1 point is earned for explanation

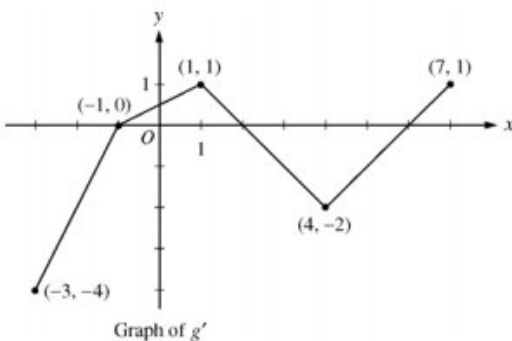
The average rate of change of f on the interval

$$-4 \leq x \leq 3 \text{ is } \frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$$

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.

2-1-1

50.



Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g' , the derivative of g , is shown above for $-3 \leq x \leq 7$.

Find the average rate of change of $g(x)$ on the interval $-3 \leq x \leq 7$.

Part C

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$

2-1-1

51.

t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/minute)	0	100	40	-120	-150

Train A runs back and forth on an east-west section of railroad track. Train A 's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

Find the average acceleration of train A over the interval $2 \leq t \leq 8$.

Part A

1 point is earned for average acceleration

$$\text{average accel} = \frac{V_A(8) - V_A(2)}{8 - 2} = \frac{-120 - 100}{6} = \frac{110}{3} m/\text{min}^2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for average acceleration

$$\text{average accel} = \frac{V_A(8) - V_A(2)}{8 - 2} = \frac{-120 - 100}{6} = \frac{110}{3} m/\text{min}^2$$

2-1-1

52.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

Explain why there must be a value c for $1 < c < 3$ such that $h'(c) = -5$.

Part B

1 point is earned for: $\frac{h(3)-h(1)}{3-1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3)-h(1)}{3-1} = \frac{-7-3}{3-1} = -5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c , $1 < c < 3$, such that $h'(c) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{h(3)-h(1)}{3-1}$

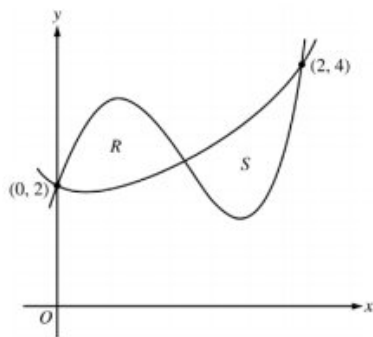
1 point is earned for the conclusion, using MVT

$$\frac{h(3)-h(1)}{3-1} = \frac{-7-3}{3-1} = -5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c , $1 < c < 3$, such that $h'(c) = -5$.

2-1-1

53.



Let f and g be the functions defined by $f(x) = 1 + x + e^{x^2-2x}$ and $g(x) = x^4 - 6.5x^2 + 6x + 2$. Let R and S be the two regions enclosed by the graphs of f and g shown in the figure above.



Let h be the vertical distance between the graphs of f and g in region S . Find the rate at which h changes with respect to x when $x = 1.8$.

Part C

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$

2-1-1

54. Let f be a function that is everywhere differentiable and that has the following properties.

(i) $f(x+h) = \frac{f(x)+f(h)}{f(-x)+f(-h)}$ for all real numbers h and x (ii) $f(x) > 0$ for all real numbers x . (iii) $f'(0) = -1$.

Use the definition of the derivative to find $f'(x)$ in terms of $f(x)$.

Part D

1 point is earned for defn of derivative

1 point is earned if invokes (c)

1 point is earned if gets $f(0)$ in context of a solution

1 point is earned if invokes (iii) and solves

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h} \\ &= f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} \\ &= f(x)f'(0) = -f(x) \end{aligned}$$

✓

0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for defn of derivative

1 point is earned if invokes (c)

1 point is earned if gets $f(0)$ in context of a solution

1 point is earned if invokes (iii) and solves

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h} \\ &= f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} \\ &= f(x)f'(0) = -f(x) \end{aligned}$$

55. $\lim_{h \rightarrow 0} \frac{\ln(4+h) - \ln(4)}{h}$ is

(A) 0

(B) $\frac{1}{4}$

(C) 1

(D) e

(E) nonexistent

✓

2-1-1

56. If $f(x) = 3x^2 + 2x$, then $f'(x) =$

(A) $\lim_{h \rightarrow 0} \frac{(3x^2 + 2x + h) - (3x^2 + 2x)}{h}$

(B) $\lim_{x \rightarrow 0} \frac{(3x^2 + 2x + h) - (3x^2 + 2x)}{h}$

(C) $\lim_{h \rightarrow 0} \frac{(3(x+h)^2 + 2(x+h)) - (3x^2 + 2x)}{h}$ ✓

(D) $\lim_{x \rightarrow 0} \frac{(3(x+h)^2 + 2(x+h)) - (3x^2 + 2x)}{h}$

57. if $f(x) = (x^2 + 1)^3$, what is $\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x + 1}$?

(A) -24 ✓

(B) -8

(C) 0

(D) 12

58. $\lim_{x \rightarrow e} \frac{(x^{20} - 3x) - (e^{20} - 3e)}{x - e}$ is

(A) 0

(B) $20e^{19} - 3$ ✓

(C) $e^{20} - 3e$

(D) nonexistent

Answer B

This option is correct. The limit of this difference quotient is of the form $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ where $f(x) = x^{20} - 3x$ and $a = e$. This is one way to express the derivative of f at $x = e$.

$$f'(x) = d/dx (x^{20} - 3x) = 20x^{19} - 3$$

$$\therefore f'(e) = 20e^{19} - 3$$

2-1-1

59.

t (minutes)	0	4	8	12	16
$H(t)$ ($^{\circ}\text{C}$)	65	68	73	80	91

The temperature, in degrees Celsius ($^{\circ}\text{C}$), of an oven being heated is modeled by an increasing differentiable function H of time t , where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.

Use the data in the table to estimate the instantaneous rate at which the temperature of the oven is changing at time $t = 10$. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for difference quotient

1 point is earned for answer with units

$$H'(10) = \frac{H(12) - H(8)}{12 - 8} = \frac{80 - 73}{4} = \frac{7}{4}^{\circ}\text{C}/\text{min}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for difference quotient

1 point is earned for answer with units

$$H'(10) = \frac{H(12) - H(8)}{12 - 8} = \frac{80 - 73}{4} = \frac{7}{4}^{\circ}\text{C}/\text{min}$$

60. Let f be the function given by $f(x) = x^3 - 6x^2 + 8x - 2$. What is the instantaneous rate of change of f at $x = 3$?

(A) -5

(B) $-\frac{15}{4}$

(C) -1

(D) 6

(E) 17

✓

2-1-1

61.

t (hours)	0	1	2	3	4	5	6
$s(t)$ (miles)	0	25	55	92	150	210	275

The table above gives the distance $s(t)$, in miles, that a car has traveled at various times t , in hours, during a 6-hour trip. The graph of the function s is increasing and concave up. Based on the information, which of the following could be the velocity of the car, in miles per hour, at time $t = 3$?

(A) 37

(B) 49

(C) 58

(D) 65

(E) 92



2-1-1

62. Let f be a function that is everywhere differentiable and that has the following properties.

(i) $f(x + h) = \frac{f(x) + f(h)}{f(-x) + f(-h)}$ for all real numbers h and x .

(ii) $f(x) > 0$ for all real numbers x .

(iii) $f'(0) = -1$.

(a) Find the value of $f(0)$.

(b) Show that $f(-x) = \frac{1}{f(x)}$ for all real numbers x .

(c) Using part (b), show that $f(x + h) = f(x)f(h)$ for all real numbers h and x .

(d) Use the definition of the derivative to find $f'(x)$ in terms of $f(x)$.

Part A

1 point is earned for solution

$$\text{Let } x = h = 0. \text{ Then } f(0) = f(0 + 0) = \frac{f(0) + f(0)}{f(0) + f(0)} = 1$$



0

1

The student response earns all of the following points:

1 point is earned for solution

$$\text{Let } x = h = 0. \text{ Then } f(0) = f(0 + 0) = \frac{f(0) + f(0)}{f(0) + f(0)} = 1$$

Part B

1 point is earned if uses $h = 0$ or other substitution in (i)

1 point is earned for solving

$$\text{Let } h = 0 \quad f(x + 0) = f(x) = \frac{f(x) + f(0)}{f(-x) + f(-0)}$$

$$\text{Use } f(0) = 1 \text{ and solve for } f(x) = \frac{1}{f(-x)}$$

OR

$$\text{Note that } f(-x + 0) = \frac{f(-x) + f(0)}{f(x) + f(0)} \text{ is the reciprocal of } f(x).$$

2-1-1



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if uses $h = 0$ or other substitution in (i)

1 point is earned for solving

$$\text{Let } h = 0 \quad f(x + 0) = f(x) = \frac{f(x) + f(0)}{f(-x) + f(-0)}$$

$$\text{Use } f(0) = 1 \text{ and solve for } f(x) = \frac{1}{f(-x)}$$

OR

$$\text{Note that } f(-x + 0) = \frac{f(-x) + f(0)}{f(x) + f(0)} \text{ is the reciprocal of } f(x).$$

Part C

1 point is earned if invokes result of (b) in (i)

1 point is earned for solving

$$\begin{aligned} f(x + h) &= \frac{f(x) + f(h)}{\frac{1}{f(x)} + \frac{1}{f(h)}} \\ &= \frac{f(x) + f(h)}{f(h) + f(x)} f(x) f(h) \\ &= f(x) f(h) \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if invokes result of (b) in (i)

1 point is earned for solving

2-1-1

$$\begin{aligned}
 f(x+h) &= \frac{f(x)+f(h)}{\frac{1}{f(x)}+\frac{1}{f(h)}} \\
 &= \frac{f(x)+f(h)}{f(h)+f(x)} f(x)f(h) \\
 &= f(x)f(h)
 \end{aligned}$$

Part D

1 point is earned for defn of derivative

1 point is earned if invokes (c)

1 point is earned if gets $f(0)$ in context of a solution

1 point is earned if invokes (iii) and solves

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x)f(h)-f(x)}{h} \\
 &= f(x) \lim_{h \rightarrow 0} \frac{f(h)-1}{h} \\
 &= f(x)f'(0) = -f(x)
 \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for defn of derivative

1 point is earned if invokes (c)

1 point is earned if gets $f(0)$ in context of a solution

1 point is earned if invokes (iii) and solves

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x)f(h)-f(x)}{h} \\
 &= f(x) \lim_{h \rightarrow 0} \frac{f(h)-1}{h} \\
 &= f(x)f'(0) = -f(x)
 \end{aligned}$$

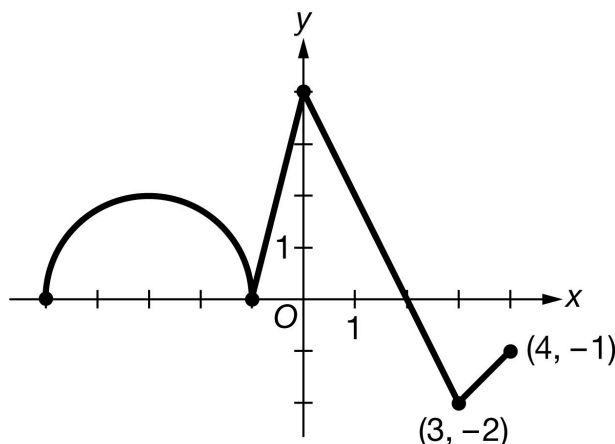
2-1-1

63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

Let g be a function that is defined on the closed interval $-5 \leq x \leq 4$ with $g(0) = 5$. The graph of g' , the derivative of g , consists of a semicircle and three line segments, as shown above.

- On what intervals, if any, is g decreasing? Give a reason for your answer.
- Find the average rate of change of g on the interval $[-3, 3]$. Is there a value c , $-3 < c < 3$, for which $g'(c)$ is equal to the average rate of change of g on the interval $[-3, 3]$? Justify your answer.
- Find the x -coordinate of each point of inflection of the graph of g on the open interval $(-5, 4)$. Justify your answer.
- Let $H(x) = e^{3x}g(x)$. Find the value of $H'(0)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not have to say that g is continuous to earn the point.

A response with correct endpoints can use open, closed, or half-open interval notation to earn the point.



0

1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

2-1-1

g is decreasing on the interval $[2, 4]$ because g is continuous and $g'(x) < 0$ on $(2, 4]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for declaring that the average rate of change is either $\frac{g(3)-g(-3)}{3-(-3)}$ or $\frac{\int_{-3}^3 g'(x) dx}{6}$. The difference quotient and the definite integral must contain $\frac{1}{6}$ to earn the point.

The second point is earned for finding the value of the average rate of change, $\frac{\pi+5}{6}$ or equivalent.

The third point is for stating that g is continuous because g is differentiable (on $[-3, 3]$). Stating that g is both differentiable and continuous on the interval is not enough to earn this point.

The fourth point is for applying either the Mean Value Theorem (MVT) and concluding “yes,” or for applying the Intermediate Value Theorem (IVT) and concluding “yes.” Neither theorem needs to be named if the appropriate conditions are checked and the correct conclusion is reached.

If a response does not appropriately cite/apply the MVT or IVT, the response does not earn the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Difference quotient or definite integral
- ☐ Average rate of change
- ☐ Differentiable implies continuous
- ☐ Conclusion using Mean Value Theorem
– OR –
Conclusion using Intermediate Value Theorem

Solution:

$$\begin{aligned}\text{Average rate of change} &= \frac{g(3)-g(-3)}{3-(-3)} = \frac{\int_{-3}^3 g'(x) dx}{6} \\ &= \frac{\pi+6-1}{6} = \frac{\pi+5}{6}\end{aligned}$$

g is differentiable on $[-3, 3]$. $\Rightarrow g$ is continuous on $[-3, 3]$.

The Mean Value Theorem guarantees that there exists a value of c in the interval $(-3, 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3, 3]$.

– OR –

g' is continuous on $[-3, 3]$.

2-1-1

The range of g' on the interval $[-3, 3]$ is $-2 \leq g'(x) \leq 4$. Because $-2 < \frac{\pi+5}{6} < 4$, there exists a value of c in the interval $(-3, 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3, 3]$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must identify all four answers.

A response that provides a correct identification and justification for just two (or three) of the four answers earns 1 of the 2 points.

The second point can be earned by correctly discussing the signs of the slopes of the graph of g' .

The second point cannot be earned by the use of vague or undefined terms such as “it” or “the graph” or “the derivative.”

The second point cannot be earned by only discussing the concavity of the graph of g .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answers
- ☐ Justification

Solution:

The graph of g will have a point of inflection where the graph of g' changes from increasing to decreasing or from decreasing to increasing. This occurs at $x = -3$, $x = -1$, $x = 0$, and $x = 3$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for using the product rule and chain rule to correctly find $H'(x)$ or $H'(0)$.

The second point is earned for correctly using the values of $g(0)$ and $g'(0)$ in the calculation of $H'(0)$. This point can be earned without the first point if there is a single error in finding $H'(x)$.

A response of $3 \cdot 5 + 4$ is the minimal response that earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $H'(x)$
- ☐ Answer

Solution:

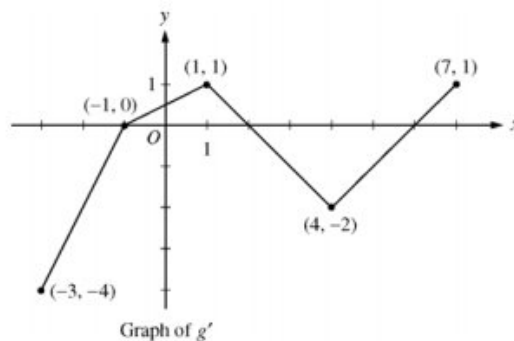
2-1-1

$$H'(x) = 3e^{3x} \cdot g(x) + e^{3x} \cdot g'(x)$$

$$H'(0) = 3 \cdot g(0) + g'(0) = 3 \cdot 5 + 4 = 19$$

2-1-1

64.



Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g' , the derivative of g , is shown above for $-3 \leq x \leq 7$.

- Find the x -coordinate of all points of inflection of the graph of $y = g(x)$ for $-3 < x < 7$. Justify your answer.
- Find the absolute maximum value of g on the interval $-3 \leq x \leq 7$. Justify your answer.
- Find the average rate of change of $g(x)$ on the interval $-3 \leq x \leq 7$.
- Find the average rate of change of $g'(x)$ on the interval $-3 \leq x \leq 7$. Does the Mean Value Theorem applied on the interval $-3 \leq x \leq 7$ guarantee a value of c , for $-3 < c < 7$, such that $g''(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for x -values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for x -values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.

Part B

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

2-1-1

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_2^{-3} g'(x) dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$

$$g(2) = 5$$

$$g(7) = 5 + \int_2^7 g'(x) dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_2^{-3} g'(x) dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$

$$g(2) = 5$$

$$g(7) = 5 + \int_2^7 g'(x) dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.

Part C

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(-3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$



0	1	2
---	---	---

The student response earns all of the following points:

2-1-1

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(-3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$

Part D

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g'(7) - g'(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g'(7) - g'(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.

2-1-1

65.  Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.
- (a) Find the average rate of change of $A(t)$ over the interval $0 \leq t \leq 30$. Indicate units of measure.
- (b) Find the value of $A'(15)$. Using correct units, interpret the meaning of the value in the context of the problem.
- (c) Find the time t for which the amount of grass clippings in the bin is equal to the average amount of grass clippings in the bin over the interval $0 \leq t \leq 30$.
- (d) For $t > 30$, $L(t)$, the linear approximation to A at $t = 30$, is a better model for the amount of grass clippings remaining in the bin. Use $L(t)$ to predict the time at which there will be 0.5 pound of grass clippings remaining in the bin. Show the work that leads to your answer.

Part A

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$

Part B

1 point is earned for $A'(15)$

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $A'(15)$

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.

Part C

2-1-1

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or 12.414)}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or 12.414)}$$

Part D

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \Rightarrow t = 35.054$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

2-1-1

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \Rightarrow t = 35.054$$

2-1-1

66.  A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S , where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.

- (a) According to the model, what is the height of the water in the can at the end of the 60-day period?
- (b) According to the model, what is the average rate of change in the height of water in the can over the 60-day period? Show the computations that lead to your answer. Indicate units of measure.
- (c) Assuming no evaporation occurs, at what rate is the volume of water in the can changing at time $t = 7$? Indicate units of measure.
- (d) During the same 60-day period, rain on Monsoon Mountain accumulates in a can identical to the one in Stormville. The height of the water in the can on Monsoon Mountain is modeled by the function M , where $M(t) = 1/400(3t^3 - 30t^2 + 330t)$. The height $M(t)$ is measured in millimeters, and t is measured in days for $0 \leq t \leq 60$. Let $D(t) = M'(t) - S'(t)$. Apply the Intermediate Value Theorem to the function D on the interval $0 \leq t \leq 60$ to justify that there exists a time t , $0 < t < 60$, at which the heights of water in the two cans are changing at the same rate.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$S(60) = \int_0^{60} S'(t) dt = 171.813 \text{ mm}$$

Part B

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$

2-1-1



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$\frac{S(60) - S(0)}{60} = 2.863 \text{ or } 2.864 \text{ mm/day}$$

Part C

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$V(t) = 100\pi S(t)$$

$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218 \text{ mm}^3/\text{day}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for relationship between V and S

1 point is earned for the answer

1 point earned for units in (b) or (c)

$$V(t) = 100\pi S(t)$$

$$V'(7) = 100\pi S'(7) = 602.218$$

The volume of water in the can is increasing at a rate of $602.218 \text{ mm}^3/\text{day}$.

Part D

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

2-1-1

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$$D(0) = -0.675 < 0 \text{ and } D(60) = 69.37730 > 0$$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t , $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

67. The function f is not differentiable at $x = 5$. Which of the following statements must be true?

(A) f is not continuous at $x = 5$.

(B) $\lim_{x \rightarrow 5} f(x)$ does not exist.

(C) $\lim_{x \rightarrow 5} \frac{f(x) - f(5)}{x - 5}$ does not exist.



(D) $\int_0^5 f(x) \, dx$ does not exist.

Answer C

Correct. The expression $\lim_{x \rightarrow 5} \frac{f(x) - f(5)}{x - 5}$ is one form of the definition of the derivative of f at $x = 5$. Since f is not differentiable at $x = 5$, however, this limit does not exist.

2-1-1

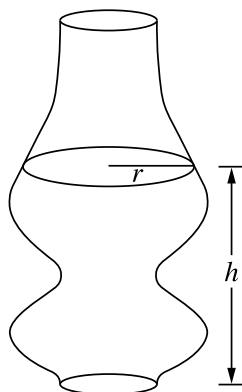
68.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



h (centimeters)	0	6.2	13	20
$r(h)$ (centimeters)	2.5	2.7	4.3	2.6

Juice is sold in 20-centimeter-tall bottles with horizontal cross sections parallel to the base that are circles, as shown in the figure. The radius of the circular cross section at height h above the base of the bottle is given by a differentiable function r , where h and $r(h)$ are measured in centimeters. Selected values of $r(h)$ are given in the table shown.

- (a) Approximate $r'(3.1)$ using the average rate of change of r over the interval $0 \leq h \leq 6.2$. Show the computations that lead to your answer. Indicate units of measure.

- (b) Interpret the meaning of $\frac{1}{20} \int_0^{20} r(h) \, dh$ in the context of the problem. Use a trapezoidal sum with the three subintervals $[0, 6.2]$, $[6.2, 13]$, and $[13, 20]$ to approximate the value of $\frac{1}{20} \int_0^{20} r(h) \, dh$.

- (c) The radius of the circular cross section at height h above the base of the bottle can also be modeled by the function f , where h and $f(h)$ are measured in centimeters. Let $f(h) = 3\sqrt{2 - 1.25 \cos\left(\frac{\pi}{100}(h - 20)^2\right)}$ for $0 \leq h \leq 20$. Based on the model, at a height of $h = 12$ centimeters, is the radius increasing or decreasing as h increases? Give a reason for your answer.

- (d) Using the model f defined in part (c), find the volume of the bottle.

Part A

2-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must provide a difference and a quotient. A response of only $\frac{2.7-2.5}{6.2}$ or $\frac{0.2}{6.2-0}$ earns the first point.

To earn the second point a response must have $\frac{\text{cm}}{\text{cm}}$ (or the equivalent) attached to a numerical value. These units do not reduce to 1 because the numerator, cm, refers to radius of the bottle and the denominator, cm, refers to height of the bottle.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Units

Solution:

$$r'(3.1) \approx \frac{r(6.2) - r(0)}{6.2 - 0} = \frac{2.7 - 2.5}{6.2}$$

0.032 centimeter per centimeter

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide “average,” “radius,” and “the interval from 0 to 20” (or equivalent such as “throughout the bottle” or “from the bottom to the top”). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the trapezoidal sum must be correct.

If there is any error in the trapezoidal sum, the response does not earn the third point.

A response of $\frac{1}{20} \left(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2} \right)$ earns the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

A response that is missing the factor of $\frac{1}{2}$ in the trapezoidal sum does not earn the second and third points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation
- ☐ Form of trapezoidal sum
- ☐ Approximation

Solution:

2-1-1

$\frac{1}{20} \int_0^{20} r(h) \, dh$ is the average value of the radius of the juice bottle, in centimeters, over the interval $0 \leq h \leq 20$ centimeters.

$$\begin{aligned} & \frac{1}{20} \int_0^{20} r(h) \, dh \\ & \approx \frac{1}{20} \left(6.2 \cdot \frac{r(0)+r(6.2)}{2} + 6.8 \cdot \frac{r(6.2)+r(13)}{2} + 7 \cdot \frac{r(13)+r(20)}{2} \right) \\ & = \frac{1}{20} \left(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2} \right) \\ & = \frac{1}{20} \cdot 64.07 = 3.2035 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the second point, a response does not need to provide a value for $f'(12)$; the sign of $f'(12)$ is sufficient.

Any presented value of $f'(12)$ must be correct for the digits presented, up to three decimal places.

If a response presents an incorrect sign or value for $f'(12)$, the response does not earn the second point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $f'(12) = -0.0006660598$. If a response presents this value, it earns the first point but is not eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(12)$
- ☐ Answer with reason

Solution:

$$f'(12) = -0.535902$$

At a height of 12 centimeters, the radius is decreasing as h increases because $f'(12) < 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral with an integrand of $(f(h))^2$, regardless of any subsequent errors in presentation of $f(h)$. A response that *begins* with an error in copying $f(h)$ into an integrand of the form $(f(h))^2$ does not earn the first point, but is eligible to earn the second point for a correct answer (1007.112 or 1007.111).

An answer of 320.574π rounds and truncates to 1007.028, and therefore does not earn the second point. Similarly, an answer of 320.573π truncates to 1007.109 and rounds to 1007.110, and therefore does not earn the second point. However, answers of 320.5736π or 320.5737π do round and truncate to the correct answer and earn the second point.

2-1-1

Degree mode: $\pi \int_0^{20} (f(h))^2 dh = 427.508$ (or 427.507) or 136.0799156π . A response that did not earn the second point in part (c) because of degree mode is eligible to earn the answer point in part (d) for this value. Furthermore, in degree mode 136.079π rounds to 427.505 and truncates to 427.504, and therefore does not earn the second point. However, in degree mode 136.080π is accurate to three decimals and is eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

$$\text{Volume} = \pi \int_0^{20} (f(h))^2 dh$$

$$= 1007.112 \text{ (or } 1007.111) \text{ cubic centimeters}$$

2-1-1

69.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	4	9	12
$R(t)$ (bears per year)	150	156	159	161

The rate at which the number of bears in a region is changing is modeled by the differentiable function R , where $R(t)$ is measured in bears per year and t is measured in years. Selected values of $R(t)$ are shown in the table.

(a) Approximate $R'(7)$ using the average rate of change of R over the interval $4 \leq t \leq 9$. Show the computations that lead to your answer, and indicate units of measure.

(b) Interpret the meaning of $\frac{1}{12} \int_0^{12} R(t) \, dt$ in the context of the problem. Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\frac{1}{12} \int_0^{12} R(t) \, dt$.

(c) The rate at which the number of bears in the region is changing can also be modeled by the function P defined by $P(t) = 150e^{0.02\sqrt{t}}$, where $P(t)$ is measured in bears per year and t is measured in years. There are 300 bears in the region at time $t = 0$. Use this model to find the number of bears in the region at time $t = 12$. Show the setup for your calculations, and give your answer to the nearest whole number of bears.

(d) Using the model defined in part (c), find the time t , for $0 \leq t \leq 12$, at which the number of bears in the region is a maximum. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must provide a numerical difference and a quotient. A response of only or earns the first point.

To earn the second point, a response must have “bear per year per year” (or the equivalent) attached to a numerical value.



0	1	2
---	---	---

2-1-1

The student response accurately includes **both** of the criteria below.

- ☐ Approximation with supporting work
- ☐ Units

Solution:

$$R'(7) \approx \frac{R(9)-R(4)}{9-4} = \frac{159-156}{5} = 0.6$$

$$R'(7) \approx 0.6 \text{ bear per year per year}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide “average,” “rate of change,” and “from time $t = 0$ to $t = 12$ ” (or equivalent). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the Riemann sum must be correct. The factor of $\frac{1}{12}$ is assessed in the third point.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of only $\frac{1}{12}(150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3)$ earns the second and third points.

A response that presents the correct average value with accompanying work that does not explicitly present the six factors and/or the sum process does not earn the second point but will earn the third point. For example, responses of “ $\frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}$ ” or “600, 780, 477 $\rightarrow \frac{1857}{12}$ ” would earn the third point but not the second.

A response that presents an answer of $\frac{156 \cdot 4 + 159 \cdot 5 + 161 \cdot 3}{12}$ (which uses a right Riemann sum) earns 1 of the last 2 points. If this response is simplified, the result must be $\frac{1902}{12}$, $\frac{951}{6}$, $\frac{317}{2}$, or 158.5.

An unsupported response of $\frac{1857}{12}$, $\frac{619}{4}$, or 154.75 does not earn the second or third point.

A response that miscommunicates by equating two unequal quantities, e.g., $\int_0^{12} R(t) dt = \frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}$, does not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation
- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

2-1-1

$\frac{1}{12} \int_0^{12} R(t) \, dt$ is the average rate of change in the number of bears in the region, in bears per year, from time $t = 0$ to time $t = 12$ years.

$$\begin{aligned} \int_0^{12} R(t) \, dt &\approx R(0) \cdot (4 - 0) + R(4) \cdot (9 - 4) + R(9) \cdot (12 - 9) \\ &= 150 \cdot (4 - 0) + 159 \cdot (9 - 4) + 159 \cdot (12 - 9) \\ &= 150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3 = 1857 \end{aligned}$$

$$\frac{1}{12} \int_0^{12} R(t) \, dt \approx \frac{1857}{12} = \frac{619}{4} = 154.75 \text{ bears per year}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral $\int_a^b P(t) \, dt$ with any limits of integration. If the limits are incorrect, the response does not earn the second point.

The second point cannot be earned without the first point, and is earned only for an answer of 2185 or 2185.339 or 2185.338.

Missing differentials:

- A response of only $\int_0^{12} P(t)$ or $\int_0^{12} P(t) = 1885.3389$ earns the first point but not the second.
- A response of $300 + \int_0^{12} P(t)$ earns the first point and in the presence of the correct answer earns the second point.
- A response of $\int_0^{12} P(t) + 300$ does not earn the first point but earns the second point for the correct answer.

A response that miscommunicates, e.g., $\int_0^{12} P(t) \, dt = 1885.3389 + 300$, does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

Solution:

$$300 + \int_0^{12} P(t) \, dt = 300 + 1885.3389 = 2185.3389$$

At $t = 12$ years, there are 2185 bears in the region.

Part D

2-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers $P = 0$ earns the first point.

A response of “at time $t = 12$ ” with no justification does not earn the second point.

The second point cannot be earned for any answer other than time $t = 12$.

Alternate solution using Candidates Test:

$P(t) = 0$ has no solutions in the interval $0 \leq t \leq 12$, so the maximum number of bears occurs at an endpoint of the interval.

The region has 300 bears at time $t = 0$.

From part (c), the region has 2185 bears at time $t = 12$.

For $0 \leq t \leq 12$, the maximum number of bears is at time $t = 12$ years.

- The scoring is the same as the model solution.
- A response using this argument may import an incorrect value for the number of bears in the region in year 12 from part (c), provided that value is larger than 300.
- A response using this argument need not report the number of bears at time $t = 12$. Stating that this number is greater than 300 is sufficient.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers sign of P
- ☐ Answer with justification

Solution:

$P(t) > 0$ for all $t \geq 0$, so the bear population is always increasing.

For $0 \leq t \leq 12$, the maximum number of bears is at time $t = 12$ years.

2-1-1

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	15	45	70	90	100
$A(t)$ (automobiles)	0	30	190	250	405	600

Prior to a sporting event, the number of automobiles that have entered a stadium parking lot is modeled by the differentiable function A , where t is the number of minutes since the parking lot opened. Values of $A(t)$ for selected values of t are given in the table above.

According to the model, what is the average rate at which automobiles enter the parking lot, in automobiles per minute, over the time interval $45 \leq t \leq 90$ minutes?

Write $A'(95)$ as the limit of a difference quotient. Use the data in the table to approximate $A'(95)$. Show the computations that lead to your answer.

What is the shortest time interval during which it is guaranteed that $A(t) = 400$ for some time t in the interval? Justify your answer.

For $0 \leq t \leq 45$, the function f defined by $f(t) = \frac{2}{27}t^2 + \frac{8}{9}t$ models the number of automobiles that have entered the parking lot, where t is the number of minutes since the parking lot opened. Find $f'(10)$, the rate at which automobiles enter the parking lot, in automobiles per minute, at time $t = 10$ minutes.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Solution:

$$\frac{A(90) - A(45)}{90 - 45} = \frac{405 - 190}{90 - 45} = \frac{215}{45} = \frac{43}{9}$$

The average rate at which automobiles enter the parking the time interval $45 \leq t \leq 90$ minutes is $\frac{43}{9}$ automobiles per minute.

Part B

2-1-1

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

Solution:

$$A'(95) = \lim_{t \rightarrow 95} \frac{A(t) - A(95)}{t - 95}$$

$$\text{OR } A'(95) = \lim_{h \rightarrow 0} \frac{A(95+h) - A(95)}{h}$$

$$A'(95) \approx \frac{A(100) - A(90)}{100 - 90} = \frac{600 - 405}{100 - 90} = \frac{195}{10} = \frac{39}{2}$$

Part C

1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ time interval
- ☐ justification, using Intermediate Value Theorem

Solution:

The shortest time interval for which it is guaranteed that $A(t) = 400$ is $70 \leq t \leq 90$.

Because A is differentiable, A is continuous.

$$A(70) = 250 < 400 < 405 = A(90)$$

By the Intermediate Value Theorem, there must be at least one time t , for $70 \leq t \leq 90$, such that $A(t) = 400$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

2-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $f'(t)$

☐ $f'(10)$

Solution:

$$f'(t) = \frac{4}{27}t + \frac{8}{9}$$

$$f'(10) = \frac{4}{27} \cdot 10 + \frac{8}{9} = \frac{40}{27} + \frac{24}{27} = \frac{64}{27}$$

At time $t = 10$ minutes, automobiles enter the parking lot at a rate of $\frac{64}{27}$ automobiles per minute.

2-1-1

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(t) = \begin{cases} g(t) & \text{for } 0 \leq t \leq 12 \\ \frac{t^2}{8} - 3t + 83 & \text{for } 12 < t \leq 24 \end{cases}$$

t (hours)	0	4	6	8	12
$g(t)$ (degrees Fahrenheit)	87	81.5	76	70.5	65

The temperature of a room, in degrees Fahrenheit, on a certain day is modeled by the function f defined above, where g is a differentiable function and t is measured in hours. Values of $g(t)$ at selected values of t are given in the table above.

- According to the model f , what is the average rate of change of the temperature of the room over the time interval $0 \leq t \leq 12$ hours?
- Use the data in the table to approximate $f'(10)$, the instantaneous rate of change in the temperature of the room, in degrees Fahrenheit per hour, at time $t = 10$ hours. Show the computations that lead to your answer.
- Is f continuous for $0 \leq t \leq 24$? Justify your answer.
- Find $f'(20)$, the instantaneous rate of change in the temperature of the room, in degrees Fahrenheit per hour, at time $t = 20$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Solution:

$$\frac{f(12)-f(0)}{12-0} = \frac{g(12)-g(0)}{12-0} = \frac{65-87}{12} = -\frac{11}{6}$$

The average rate of change of the temperature of the room over the time interval $0 \leq t \leq 12$ hours is $-\frac{11}{6}$ degrees Fahrenheit per hour

Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

2-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation of $f'(10)$

Solution:

$$f'(10) \approx \frac{g(12) - g(8)}{12 - 8} = \frac{65 - 70.5}{4} = -\frac{5.5}{4}$$

Part C

The response must explicitly connect differentiability and continuity to earn the first point and use the fact that f is a polynomial on $12 < t \leq 24$ to explain why f is continuous on that interval to earn the second point. The third point is for continuity at $t = 12$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ differentiability of g implies continuity
- ☐ f is continuous on $12 < t \leq 24$
- ☐ $\lim_{t \rightarrow 12^-} f(t) = \lim_{t \rightarrow 12^+} f(t) = f(12)$

Solution:

Because g is differentiable, g is continuous on $0 \leq t \leq 12$. $f(t) = g(t)$ for $0 \leq t \leq 12$.

Because f is a polynomial function on $12 < t \leq 24$, f is continuous on $12 < t \leq 24$.

$$\lim_{t \rightarrow 12^-} f(t) = f(12) = g(12) = 65$$

$$\lim_{t \rightarrow 12^+} f(t) = \frac{12^2}{8} - 3(12) + 83 = 65$$

Because $\lim_{t \rightarrow 12^-} f(t) = \lim_{t \rightarrow 12^+} f(t) = f(12)$, f is continuous at $t = 12$.

Thus, f is continuous on $0 \leq t \leq 24$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $f'(t)$

☐ $f'(20)$

Solution:

$$f(t) = \frac{t^2}{8} - 3t + 83$$

$$f'(t) = \frac{1}{4}t - 3$$

$$f'(20) = \frac{1}{4}(20) - 3 = 5 - 3 = 2$$

The instantaneous rate of change in the temperature of the room at $t = 20$ hours is 2 degrees Fahrenheit per hour.

2-1-1

72. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(t) = \begin{cases} 48t + t^2 - \frac{t^3}{12} & \text{for } 0 \leq t < 6 \\ g(t) & \text{for } 6 \leq t \leq 12 \end{cases}$$

t (hours)	6	8	10	12
$g(t)$ (cubic meters)	306	376	428	474

At an excavation site, the amount of dirt that has been removed, in cubic meters, is modeled by the function f defined above, where g is a differentiable function and t is measured in hours. Values of $g(t)$ at selected values of t are given in the table above.

- According to the model f , what is the average rate of change of the amount of dirt removed over the time interval $6 \leq t \leq 12$ hours?
- Use the data in the table to approximate $f'(9)$, the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time $t = 9$ hours. Show the computations that lead to your answer.
- Is f continuous for $0 \leq t \leq 12$? Justify your answer.
- Find $f'(2)$, the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time $t = 2$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Solution:

$$\frac{f(12) - f(6)}{12 - 6} = \frac{g(12) - g(6)}{12 - 6} = \frac{474 - 306}{12 - 6} = \frac{168}{6} = 28$$

The average rate of change of the amount of dirt removed over the time interval $6 \leq t \leq 12$ hours is 28 cubic meters per hour.

Part B

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

2-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation of $f'(9)$.

Solution:

$$f'(9) \approx \frac{g(10)-g(8)}{10-8} = \frac{428-376}{10-8} = \frac{52}{2} = 26$$

Part C

The response must explicitly connect differentiability and continuity to earn the first point and use the fact that f is a polynomial for $0 \leq t < 6$ to explain why f is continuous on that interval to earn the second point. The third point is for continuity at $t = 6$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ f is continuous for $0 \leq t < 6$
- ☐ differentiability of g implies continuity
- ☐ $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$

Solution:

Because f is a polynomial function for $0 \leq t < 6$, f is continuous for $0 \leq t < 6$.

Because g is differentiable, g is continuous on $6 \leq t \leq 12$. $f(t) = g(t)$ for $6 \leq t \leq 12$.

$$\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^-} \left(48t + t^2 - \frac{t^3}{12} \right) = 48(6) + 6^2 - \frac{6^3}{12} = 288 + 36 - 18 = 306$$

$$\lim_{t \rightarrow 6^+} f(t) = f(6) = g(6) = 306$$

Because $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$, f is continuous at $t = 6$.

Thus, f is continuous for $0 \leq t \leq 12$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$, so

$f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4}$ earns both points. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $f'(t)$

☐ $f'(2)$

Solution:

$$f(t) = 48t + t^2 - \frac{t^3}{12} \text{ for } 0 \leq t < 6$$

$$f'(t) = 48 + 2t - \frac{t^2}{4} \text{ for } 0 < t < 6$$

$$f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4} = 51$$

The instantaneous rate of change in the amount of dirt removed at $t = 2$ hours is 51 cubic meters per hour.

2-1-1

73. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	3	6	8
$P(t)$ (passengers)	0	35	204	600	728

The number of passengers who have boarded a ship is modeled by the differentiable function P , where t is the number of hours since boarding began. Values of $P(t)$ for selected values of t are given in the table above.

- According to the model, what is the average rate at which passengers board the ship, in passengers per hour, over the time interval $1 \leq t \leq 8$ hours?
- Write $P'(4.5)$ as the limit of a difference quotient. Use the data in the table to approximate $P'(4.5)$. Show the computations that lead to your answer.
- Must there be a time t , for $3 \leq t \leq 6$, at which $P(t) = 500$? Justify your answer.
- The total number of gallons of water used by the passengers on the ship is modeled by the function G defined by $G(t) = 120t\sqrt{t}$ for $0 \leq t \leq 8$, where t is the number of hours since boarding began. Find $G'(4)$, the rate at which passengers use water, in gallons per hour, at time $t = 4$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Solution:

$$\frac{P(8) - P(1)}{8 - 1} = \frac{728 - 35}{7} = 99$$

The average rate at which passengers board the ship over the time interval $1 \leq t \leq 8$ hours is 99 passengers per hour.

Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-1-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

Solution:

$$P'(4.5) = \lim_{t \rightarrow 4.5} \frac{P(t) - P(4.5)}{t - 4.5}$$

OR

$$P'(4.5) = \lim_{h \rightarrow 0} \frac{P(4.5+h) - P(4.5)}{h}$$

$$P'(4.5) \approx \frac{P(6) - P(3)}{6 - 3} = \frac{600 - 204}{3} = 132$$

Part C

1 out of 2 points if correct justification does not specifically reference IVT .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ P is continuous with reason
- ☐ justification, using Intermediate Value Theorem

Solution:

Because P is differentiable, P is continuous.

$$P(3) = 204 < 500 < 600 = P(6)$$

By the Intermediate Value Theorem, there must be at least one time t , for $3 \leq t \leq 6$, such that

$$P(t) = 500.$$

Part D

The first point is earned for demonstration of a correct derivative for G and does not require a separate statement of $G'(t)$.

The derivative and evaluation of the derivative can be expressed in one line to earn both points.

2-1-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $G'(t)$

☐ $G'(4)$

Solution:

$$G(t) = 120t\sqrt{t} = 120t^{\frac{3}{2}}$$

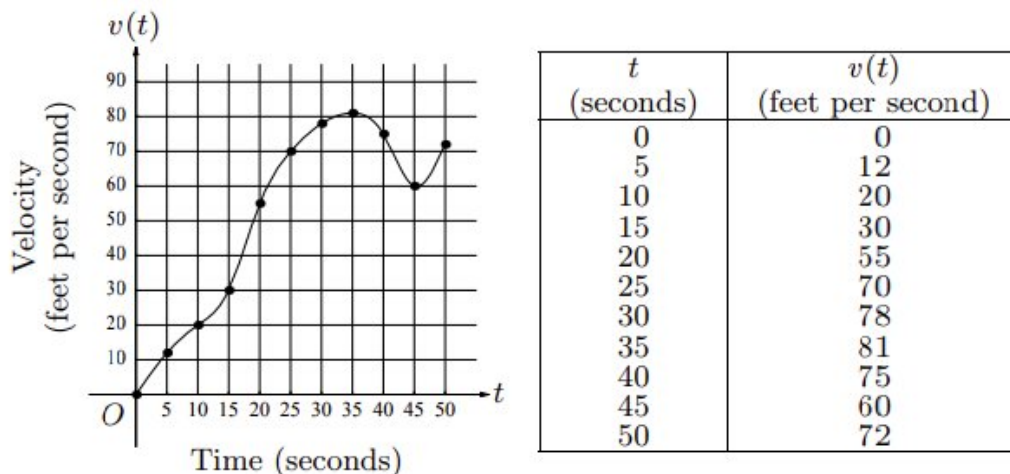
$$G'(t) = 120 \cdot \frac{3}{2}t^{\frac{1}{2}} = 180\sqrt{t}$$

$$G'(4) = 180\sqrt{4} = 180 \cdot 2 = 360$$

At time $t = 4$ hours, passengers use water at a rate of 360 gallons per hour.

2-1-1

74.



The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph.

- (a) During what intervals of time is the acceleration of the car positive? Give a reason for your answer.
- (b) Find the average acceleration of the car, in ft/sec^2 , over the interval $0 \leq t \leq 50$.
- (c) Find one approximation for the acceleration of the car, in ft/sec^2 , at $t = 40$. Show the computations you used to arrive at your answer.
- (d) Approximate $\int_0^{50} v(t) dt$ with a Riemann sum, using the midpoints of five subintervals of equal length. Using correct units, explain the meaning of this integral.

Part A

1 point is earned for (0, 35)

1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for (0, 35)

1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$

Part B

1 point is earned for the answer

2-1-1

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2

Part C

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

2-1-1

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

Part D

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned} & \int_0^{50} v(t) dt \\ & \approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ & = 10(12 + 30 + 70 + 81 + 60) \\ & = 2530 \text{ feet} \end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned} & \int_0^{50} v(t) dt \\ & \approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ & = 10(12 + 30 + 70 + 81 + 60) \\ & = 2530 \text{ feet} \end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.

2-1-1

75.  A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 3t^2 - 2t - 1$. The position $x(t)$ is 5 for $t = 2$.
- (a) Write a polynomial expression for the position of the particle at any time $t \geq 0$.
- (b) For what values of t , $0 \leq t \leq 3$, is the particle's instantaneous velocity the same as its average velocity on the closed interval $[0, 3]$?
- (c) Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.

Part A

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

$$\begin{aligned} x(t) &= \int v(t) dt = \int (3t^2 - 2t - 1) dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5; C = 3$$

$$x(t) = t^3 - t^2 - t + 3$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

$$\begin{aligned} x(t) &= \int v(t) dt = \int (3t^2 - 2t - 1) dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5; C = 3$$

$$x(t) = t^3 - t^2 - t + 3$$

2-1-1**Part B**

1 point is earned for the correct average velocity = 5

$$\begin{aligned}\text{avg. vel.} &= \frac{x(3)-x(0)}{3-0} \\ &= \frac{18-3}{3} = 5\end{aligned}$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 = \text{an avg. velocity in } [0, 3]$

$$t = \frac{1+\sqrt{19}}{3} \text{ or } 1.786$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct average velocity = 5

$$\begin{aligned}\text{avg. vel.} &= \frac{x(3)-x(0)}{3-0} \\ &= \frac{18-3}{3} = 5\end{aligned}$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 = \text{an avg. velocity in } [0, 3]$

$$t = \frac{1+\sqrt{19}}{3} \text{ or } 1.786$$

Part C

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

2-1-1

$$\begin{aligned}\text{distance} &= \int_0^3 |v(t)| dt \\ &= \int_0^3 |3t^2 - 2t - 1| dt = 17\end{aligned}$$

or

$$\begin{aligned}v(t) &= 3t^2 - 2t - 1 = 0 \\ t &= -\frac{1}{3}, t = 1 \\ x(0) &= 3 \\ x(1) &= 1 - 1 - 1 + 3 = 2 \\ x(3) &= 27 - 9 - 3 + 3 = 18\end{aligned}$$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

$$\text{distance} = (3 - 2) + (18 - 2) = 17$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

$$\begin{aligned}\text{distance} &= \int_0^3 |v(t)| dt \\ &= \int_0^3 |3t^2 - 2t - 1| dt = 17\end{aligned}$$

or

$$\begin{aligned}v(t) &= 3t^2 - 2t - 1 = 0 \\ t &= -\frac{1}{3}, t = 1 \\ x(0) &= 3 \\ x(1) &= 1 - 1 - 1 + 3 = 2 \\ x(3) &= 27 - 9 - 3 + 3 = 18\end{aligned}$$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

$$\text{distance} = (3 - 2) + (18 - 2) = 17$$

2-1-1

76.

$$g(x) = \begin{cases} x^2 + 1 & \text{for } x < 3 \\ 13 - x & \text{for } x \geq 3 \end{cases}$$

Let g be the function given above. What is the average rate of change of g on the closed interval $[1, 5]$?

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) 5

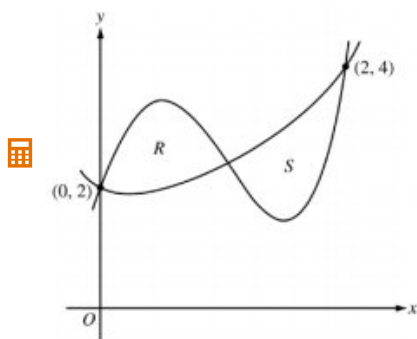
(D) Undefined

**Answer B**

Correct. The average rate of change of g on the closed interval $[1, 5]$ is $\frac{g(5)-g(1)}{5-1}$. For this piecewise defined function, $g(1) = (x^2 + 1)|_{x=1} = 2$ and $g(5) = (13 - x)|_{x=5} = 8$. Therefore the average rate of change is $\frac{g(5)-g(1)}{5-1} = \frac{8-2}{4} = \frac{6}{4} = \frac{3}{2}$.

2-1-1

77.



Let f and g be the functions defined by $f(x) = 1 + x + e^{x^2-2x}$ and $g(x) = x^4 - 6.5x^2 + 6x + 2$. Let R and S be the two regions enclosed by the graphs of f and g shown in the figure above.

(a) Find the sum of the areas of regions R and S .

(b) Region S is the base of a solid whose cross sections perpendicular to the x -axis are squares. Find the volume of the solid.

(c) Let h be the vertical distance between the graphs of f and g in region S . Find the rate at which h changes with respect to x when $x = 1.8$.

Part A

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y = f(x)$ and $y = g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and

$(A, B) = (1.032832, 2.401108)$.

$$\begin{aligned} \text{Area} &= \int_0^A [g(x) - f(x)]dx + \int_A^2 [f(x) - g(x)]dx \\ &= 0.997427 + 1.006919 = 2.004 \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y = f(x)$ and $y = g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and

$(A, B) = (1.032832, 2.401108)$.

$$\begin{aligned} \text{Area} &= \int_0^A [g(x) - f(x)]dx + \int_A^2 [f(x) - g(x)]dx \\ &= 0.997427 + 1.006919 = 2.004 \end{aligned}$$

2-1-1

Part B

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$

Part C

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$

2-1-1

78. The function f is given by $f(x) = 1 + 2 \sin x$. What is the average rate of change of f over the interval $[0, \frac{\pi}{2}]$?

- (A) $\frac{4}{\pi}$
(B) $\frac{6}{\pi}$
(C) $\frac{8}{\pi}$
(D) 2

**Answer A**

Correct. The difference quotient $\frac{f(\frac{\pi}{2}) - f(0)}{\frac{\pi}{2} - 0}$ is the average rate of change of f over the interval $[0, \frac{\pi}{2}]$.

$$\frac{f(\frac{\pi}{2}) - f(0)}{\frac{\pi}{2} - 0} = \frac{(1 + 2 \sin \frac{\pi}{2}) - (1 + 2 \sin 0)}{\frac{\pi}{2}} = \frac{(1 + 2(1)) - (1 + 2(0))}{\frac{\pi}{2}} = \frac{2}{\frac{\pi}{2}} = \frac{4}{\pi}$$

79. Let f be the function defined by $f(x) = \sec x + \csc x$. Which of the following expressions is the average rate of change of f over the interval $[\frac{\pi}{4}, \frac{3\pi}{8}]$?

- (A) $\frac{f(\frac{3\pi}{8}) + f(\frac{\pi}{4})}{2}$
(B) $\frac{f(\frac{3\pi}{8}) + f(\frac{\pi}{4})}{\frac{3\pi}{8} + \frac{\pi}{4}}$
(C) $\frac{f(\frac{3\pi}{8}) - f(\frac{\pi}{4})}{\frac{3\pi}{8} - \frac{\pi}{4}}$
(D) $f(\frac{3\pi}{8}) - f(\frac{\pi}{4})$

**Answer C**

Correct. The average rate of change of a function f over an interval $[a, b]$ is given by the difference quotient $\frac{f(b) - f(a)}{b - a}$.

80. Let f be the function defined by $f(x) = 2 \sin x + \cos x$. The average rate of change of f over the interval $[0, b]$ is 0.05, where $b > 0$. Which of the following is an equation that could be used to find the value of b ?

- (A) $f(b) = 0.05$
(B) $f(b) - f(0) = 0.05$
(C) $\frac{f(b) - f(0)}{b - 0} = 0.05$
(D) $\frac{f(b) + f(0)}{2} = 0.05$

**Answer C**

Correct. The average rate of change of f over the interval $[0, b]$ is given by $\frac{f(b) - f(0)}{b - 0}$. The equation $\frac{f(b) - f(0)}{b - 0} = 0.05$ can be solved for b using the graphing calculator.

2-1-1

81.

x	-2	-1	0	1	2
$g(x)$	-3	2	1	0	5

Selected values of a function g are shown in the table above. What is the average rate of change of g over the interval $[-2, 2]$?

(A) $\frac{2-(-2)}{5-(-3)}$

(B) $\frac{5-(-3)}{2-(-2)}$

(C) $\frac{5+(-3)}{2}$

(D) $\frac{-3+2+1+0+5}{5}$

Answer B

Correct. The average rate of change of g over the interval $[-2, 2]$ is given by the difference quotient

$$\frac{g(2)-g(-2)}{2-(-2)} = \frac{5-(-3)}{2-(-2)}.$$

82.

x	0	2	6	12	20
$f(x)$	-8	-2	0	2	18

The function f is differentiable and has values as shown in the table above. Of the following, on which interval is the average rate of change of f greatest?

(A) $[0, 2]$

(B) $[2, 6]$

(C) $[6, 12]$

(D) $[12, 20]$

Answer A

Correct. The average rate of change of f on the interval $[a, b]$ is $\frac{f(b)-f(a)}{b-a}$.

On the interval $[0, 2]$, the average rate of change is $\frac{f(2)-f(0)}{2-0} = \frac{-2-(-8)}{2} = 3$.

On the interval $[2, 6]$, the average rate of change is $\frac{f(6)-f(2)}{6-2} = \frac{0-(-2)}{4} = \frac{1}{2}$.

On the interval $[6, 12]$, the average rate of change is $\frac{f(12)-f(6)}{12-6} = \frac{2-0}{6} = \frac{1}{3}$.

On the interval $[12, 20]$, the average rate of change is $\frac{f(20)-f(12)}{20-12} = \frac{18-2}{8} = 2$.

Of these four, the average rate of change is greatest on the interval $[0, 2]$.

2-1-1

83. Let f be the function defined by $f(x) = 2x^3 - x$. Which of the following expressions is the average rate of change of f on the interval $[1, 3]$?
- (A) $\frac{f(3)+f(1)}{2}$
- (B) $\frac{f(3)+f(1)}{3+1}$
- (C) $\frac{f(3)-f(1)}{3-1}$ ✓
- (D) $f(3) - f(1)$

Answer C

Correct. The average rate of change of a function f on an interval $[a, b]$ is given by the difference quotient $\frac{f(b)-f(a)}{b-a}$.

84.

x	1	2	3	4	5
$f(x)$	2	3	5	6	14

Selected values of a function f are shown in the table above. What is the average rate of change of f over the interval $[1, 5]$?

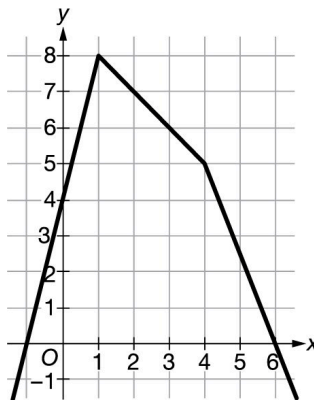
- (A) $\frac{5-1}{14-2}$
- (B) $\frac{14+2}{5+1}$
- (C) $\frac{14-2}{5-1}$ ✓
- (D) $\frac{2+3+5+6+14}{5}$

Answer C

Correct. The average rate of change over the interval $[1, 5]$ is given by the difference quotient $\frac{f(5)-f(1)}{5-1} = \frac{14-2}{5-1}$.

2-1-1

85.



The graph of the function f , shown above, consists of three line segments. What is the average rate of change of f over the interval $-1 \leq x \leq 6$?

(A) $-\frac{5}{2}$

(B) 0

(C) $\frac{1}{6}$

(D) 4

Answer B

Correct. The average rate of change of f over the interval $[-1, 6]$ is given by the difference quotient $\frac{f(6)-f(-1)}{6-(-1)} = \frac{0-0}{7} = 0$.

86. The function f is given by $f(x) = 1 + 3 \cos x$. What is the average rate of change of f over the interval $[0, \pi]$?

(A) $-\frac{6}{\pi}$

(B) $-\frac{2}{\pi}$

(C) $\frac{2}{\pi}$

(D) 1

Answer A

Correct. The difference quotient $\frac{f(\pi)-f(0)}{\pi-0}$ is the average rate of change of f over the interval $[0, \pi]$.

$$\frac{f(\pi)-f(0)}{\pi-0} = \frac{(1+3 \cos \pi)-(1+3 \cos 0)}{\pi} = \frac{1+3(-1)-1-3(1)}{\pi} = \frac{-6}{\pi}$$

2-1-1

87. Let f be the function defined by $f(x) = e^{2x}$. The average rate of change of f over the interval $[1, b]$ is 20, where $b > 1$. Which of the following is an equation that could be used to find the value of b ?

- (A) $f(b) = 20$
(B) $f(b) - f(1) = 20$

(C) $\frac{f(b)-f(1)}{b-1} = 20$

(D) $\frac{f(b)+f(1)}{2} = 20$

**Answer C**

Correct. The average rate of change of f over the interval $[1, b]$ is given by $\frac{f(b)-f(1)}{b-1}$. The equation $\frac{f(b)-f(1)}{b-1} = 20$ can be solved for b using the graphing calculator.

88.

x	-3	-1	1	3
$g(x)$	12	0	1	-4

Selected values of a function g are shown in the table above. What is the average rate of change of g over the interval $[-3, 3]$?

(A) $\frac{3-(-3)}{(-4)-12}$

(B) $\frac{(-4)-12}{3-(-3)}$

(C) $\frac{12+(-4)}{2}$

(D) $\frac{12+0+1+(-4)}{4}$

**Answer B**

Correct. The average rate of change of g over the interval $[-3, 3]$ is given by the difference quotient $\frac{g(3)-g(-3)}{3-(-3)} = \frac{(-4)-12}{3-(-3)}$.

2-1-1

89.

x	1	2	3	4	5	6
$f(x)$	-3	-1	1	2	5	10

Selected values of a function f are shown in the table above. What is the average rate of change of f over the interval $[1, 6]$?

(A) $\frac{6-1}{10-(-3)}$

(B) $\frac{10+(-3)}{6+1}$

(C) $\frac{10-(-3)}{6-1}$ ✓

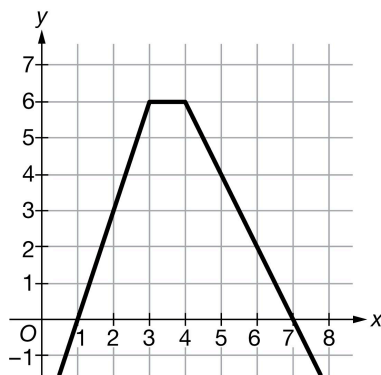
(D) $\frac{(-3)+(-1)+1+2+5+10}{6}$

Answer C

Correct. The average rate of change over the interval $[1, 6]$ is given by the difference quotient

$$\frac{f(6)-f(1)}{6-1} = \frac{10-(-3)}{6-1}.$$

90.



The graph of the function f , shown above, consists of three line segments. What is the average rate of change of f over the interval $1 \leq x \leq 7$?

(A) -2

(B) 0 ✓

(C) $\frac{1}{3}$

(D) 3

Answer B

Correct. The average rate of change of f over the interval $[1, 7]$ is given by the difference quotient $\frac{f(7)-f(1)}{7-1} = \frac{0-(-1)}{6} = \frac{1}{6}$.

2-1-1

91. Let f be a continuous function defined on the interval $0 \leq x \leq 7$. The function f and its derivatives have the properties indicated in the table shown, where DNE indicates the x -values at which the derivatives of f do not exist.

x	$x = 0$	$0 < x < 2$	$x = 2$	$2 < x < 4$	$x = 4$	$4 < x < 7$	$x = 7$
$f(x)$	-1	Negative	-4	Negative	-2	Negative	0
$f'(x)$	DNE	Negative	DNE	Positive	0	Positive	DNE
$f''(x)$	DNE	Positive	DNE	Negative	0	Positive	DNE

Part A

On what open intervals contained in $0 < x < 7$, if any, is the graph of f both increasing and concave down? Give a reason for your answer.

Part B

Find the average rate of change of f over the interval $0 \leq x \leq 4$. Can the Mean Value Theorem be applied to f on the interval $0 \leq x \leq 4$ to guarantee a value of c , for $0 < c < 4$, such that $f'(c)$ is equal to this average rate of change? Give a reason for your answer.

Part C

Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^7 f(x) dx$.

Part D

Let g be the function defined by $g(x) = x \cdot f(x)$. Is g increasing or decreasing at $x = 1$? Give a reason for your answer.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned only by an answer of $2 < x < 4$ or this interval including one or both endpoints.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” earns **P1**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” earns **P1**.

Model Solution:

f is increasing and concave down on the interval $2 < x < 4$.

2-1-1



0

1

The response accurately includes the criteria below.

- Answer

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P1** to be eligible for **P2**.
- To earn **P2**, a response must correctly cite both $f'(x) > 0$ and $f''(x) < 0$.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” does not earn **P2**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” does not earn **P2**.

Model Solution:

The graph of function f is increasing on an interval when $f'(x) > 0$ and concave down on an interval when $f''(x) < 0$.



0

1

The response accurately includes the criteria below.

- Reason

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P3**, a response must include a difference and a quotient. Any of the following is sufficient to earn **P3**:

$$\frac{f(4)-f(0)}{4-0} = -\frac{1}{4}, \frac{-2-(-1)}{4}, \text{ or } \frac{-1}{4-0}.$$

Model Solution:

$$\frac{f(4)-f(0)}{4-0} = \frac{-2-(-1)}{4} = -\frac{1}{4}$$



0

1

The response accurately includes the criteria below.

- Average rate of change

2-1-1

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response does not need to have earned **P3** to be eligible for **P4**.
- To earn **P4**, a response must note that f is not differentiable and conclude “no.”
- To earn **P4**, a response need not explicitly name the Mean Value Theorem, but if a theorem is named, it must be correct.

Model Solution:

No, the Mean Value Theorem cannot be applied to f on the interval $0 \leq x \leq 4$ because f is not differentiable at $x = 2$.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- Read “=” as “ $\approx$ ” for **P5**.
- The form of a left Riemann sum includes three terms, each of which includes a product of two factors. To earn **P5**, at least five of the six factors must be correct.
- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ earns **P5**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P5**.
- An unsupported answer of -16 does not earn **P5**.
- A completely correct right Riemann sum with accompanying work (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) earns **P5**. A response with any errors or missing factors in the right Riemann sum does not earn **P5**.

Model Solution:

$$\int_0^7 f(x)dx \approx f(0)(2-0) + f(2)(4-2) + f(4)(7-4)$$



0	1
---	---

The response accurately includes the criteria below.

- Form of left Riemann sum

Part C Point 6

2-1-1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ is eligible to earn **P6** with correct values for $f(0)$, $f(2)$, and $f(4)$ imported from the table. If any of the six factors is incorrect, the response does not earn **P6**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P6**.
- A response that includes $(-1)(2) + (-4)(2) + (-2)(3)$ or equivalent banks **P6** (i.e., subsequent errors in simplification will not be considered in scoring for **P6**).
- An unsupported answer of -16 does not earn **P6**.
- A right Riemann sum (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) does not earn **P6**.

Model Solution:

$$(-1)(2) + (-4)(2) + (-2)(3) = -16$$



0	1
---	---

The response accurately includes the criteria below.

- Answer with supporting work

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for considering either $g'(x)$ or $g'(1)$.

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
---	---

The response accurately includes the criteria below.

- Considers $g'(x)$

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned with either $g'(x) = f(x) + x \cdot f'(x)$ or $g'(1) = f(1) + 1 \cdot f'(1)$.

2-1-1

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(x)$ or $g'(1)$.

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P7** and **P8** to be eligible for **P9**.
- An eligible response that states either “decreasing because $g'(1) < 0$ ” or “decreasing because $g'(x) < 0$ for $0 < x < 2$ ” earns **P9**.
- A response that states “ $g(1)$ is decreasing” does not earn **P9**.
- A response that estimates negative values for $f(1)$ and/or $f'(1)$ using the table and reaches a correct conclusion earns **P9**.

Model Solution:

Because $g'(1) < 0$ and g' is continuous on $0 < x < 2$, g is decreasing on an interval containing $x = 1$.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

2-1-1

92.  Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by

$$F(t) = 82 + 4 \sin\left(\frac{t}{2}\right) \text{ for } 0 \leq t \leq 30,$$

where $F(t)$ is measured in cars per minute and t is measured in minutes.

- (a) To the nearest whole number, how many cars pass through the intersection over the 30-minute period?
- (b) Is the traffic flow increasing or decreasing at $t = 7$? Give a reason for your answer.
- (c) What is the average value of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.
- (d) What is the average rate of change of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$

Part B

1 point is earned for the answer with reason

$$F'(7) = -1.872 \text{ or } -1.873$$

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.



0	1
---	---

The student response earns all of the following points:

2-1-1

1 point is earned for the answer with reason

$$F'(7) = -1.872 \text{ or } -1.873$$

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.

Part C

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.899 \text{ cars/min}$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.899 \text{ cars/min}$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)

Part D

1 point is earned for the answer

$$\frac{F(15) - F(10)}{15 - 10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$

1 point is earned for units in (c) and (d)

2-1-1

Units of cars/min in (c) and cars/min² in (d)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{F(15)-F(10)}{15-10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)

2-1-1

93.

t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/minute)	0	100	40	-120	-150

Train A runs back and forth on an east-west section of railroad track. Train A 's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

- (a) Find the average acceleration of train A over the interval $2 \leq t \leq 8$.
- (b) Do the data in the table support the conclusion that train A 's velocity is -100 meters per minute at some time t with $5 < t < 8$? Give a reason for your answer.
- (c) At time $t = 2$, train A 's position is 300 meters east of the Origin Station, and the train is moving to the east. Write an expression involving an integral that gives the position of train A , in meters from the Origin Station, at time $t = 12$. Use a trapezoidal sum with three subintervals indicated by the table to approximate the position of the train at time $t = 12$.
- (d) A second train, train B , travels north from the Origin Station. At time t the velocity of train B is given by $v_B(t) = -5t^2 + 60t + 25$, and at time $t = 2$ the train is 400 meters north of the station. Find the rate, in meters per minute, at which the distance between train A and train B is changing at time $t = 2$.

Part A

1 point is earned for average acceleration

$$\text{average accel} = \frac{v_A(8) - v_A(2)}{8 - 2} = \frac{-120 - 100}{6} = -\frac{110}{3} \text{ m/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for average acceleration

$$\text{average accel} = \frac{v_A(8) - v_A(2)}{8 - 2} = \frac{-120 - 100}{6} = -\frac{110}{3} \text{ m/min}^2$$

Part B

1 point is earned for $v_A(8) < -100 < v_A(5)$

1 point is earned for the conclusion, using IVT

v_A is differentiable $\Rightarrow v_A$ is continuous

$$v_A(8) = -120 < -100 < 40 = v_A(5)$$

2-1-1

Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $v_A(t) = -100$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $v_A(8) < -100 < v_A(5)$

1 point is earned for the conclusion, using IVT

v_A is differentiable $\Rightarrow v_A$ is continuous

$$v_A(8) = -120 < -100 < 40 = v_A(5)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $5 < t < 8$, such that $v_A(t) = -100$.

Part C

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$\begin{aligned} s_A(12) &= s_A(2) + \int_2^{12} v_A(t) dt = 300 + \int_2^{12} v_A(t) dt \\ \int_2^{12} v_A(t) dt &\approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2} \\ &= -450 \\ s_A(12) &\approx 300 - 450 = -150 \end{aligned}$$

The position of Train A at time $t = 12$ minutes is approximately 150 meters west of Origin Station.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

2-1-1

$$s_A(12) = s_A(2) + \int_2^{12} v_A(t) dt = 300 + \int_2^{12} v_A(t) dt$$

$$\int_2^{12} v_A(t) dt \approx 3 \cdot \frac{100+40}{2} + 3 \cdot \frac{40-120}{2} + 4 \cdot \frac{-120-150}{2}$$

$$= -450$$

$$s_A(12) \approx 300 - 450 = -150$$

The position of Train A at time $t = 12$ minutes is approximately 150 meters west of Origin Station.

Part D

2 points are earned for implicit differentiation of distance relationship

1 point is earned for the answer

Let x be train A 's position, y train B 's position, and z the distance between train A and train B .

$$z^2 = x^2 + y^2 \Rightarrow 2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = 300, y = 400 \Rightarrow z = 500$$

$$v_B(2) = -20 + 120 + 25 = 125$$

$$500 \frac{dz}{dt} = (300)(100) + (400)(125)$$

$$\frac{dz}{dt} = \frac{80000}{500} = 160 \text{ meters per minute}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for implicit differentiation of distance relationship

1 point is earned for the answer

Let x be train A 's position, y train B 's position, and z the distance between train A and train B .

$$z^2 = x^2 + y^2 \Rightarrow 2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = 300, y = 400 \Rightarrow z = 500$$

$$v_B(2) = -20 + 120 + 25 = 125$$

$$500 \frac{dz}{dt} = (300)(100) + (400)(125)$$

$$\frac{dz}{dt} = \frac{80000}{500} = 160 \text{ meters per minute}$$

2-1-1

94. Let $F(x) = \int_0^x \sin(t^2) dt$ for $0 \leq x \leq 3$.

(a) Use the trapezoidal rule with four equal subdivisions of the closed interval $[0,1]$ to approximate $F(1)$.

(b) On what intervals is F increasing?

(c) If the average rate of change of F on the closed interval $[1,3]$ is k , find $\int_1^3 \sin(t^2) dt$ in terms of k .

Part A

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin \left(\frac{1}{4}\right)^2 + 2 \sin \left(\frac{1}{2}\right)^2 + 2 \sin \left(\frac{3}{4}\right)^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

$$\approx 0.316$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin \left(\frac{1}{4}\right)^2 + 2 \sin \left(\frac{1}{2}\right)^2 + 2 \sin \left(\frac{3}{4}\right)^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

$$\approx 0.316$$

2-1-1

Part B

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

1 point is earned for correctly analyzing sign of F' within $[0,3]$



1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$

✓

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

1 point is earned for correctly analyzing sign of F' within $[0,3]$



1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$

Part C

2-1-1

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2)dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2)dt = 2k$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2)dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2)dt = 2k$$

2-1-1

95.  The wind chill is the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v , in miles per hour (mph). If the air temperature is 32°F , then the wind chill is given by $W(v) = 55.6 - 22.1v^{0.16}$ and is valid for $5 \leq v \leq 60$.

- (a) Find $W'(20)$. Using correct units, explain the meaning of $W'(20)$ in terms of the wind chill.
- (b) Find the average rate of change of W over the interval $5 \leq v \leq 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \leq v \leq 60$.
- (c) Over the time interval $0 \leq t \leq 4$ hours, the air temperature is a constant 32°F . At time $t = 0$, the wind velocity is $v = 20$ mph. If the wind velocity increases at a constant rate of 5 mph per hour, what is the rate of change of the wind chill with respect to time at $t = 3$ hours? Indicate units of measure.

Part A

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at $0.286^{\circ}\text{F}/\text{mph}$.

1 point is earned for units in (a) and (c)

Units of $^{\circ}\text{F}/\text{mph}$ in (a) and $^{\circ}\text{F}/\text{hr}$ in (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at $0.286^{\circ}\text{F}/\text{mph}$.

1 point is earned for units in (a) and (c)

Units of $^{\circ}\text{F}/\text{mph}$ in (a) and $^{\circ}\text{F}/\text{hr}$ in (c)

Part B

1 point is earned for the average rate of change.

1 point is earned for $W'(v) = \text{average rate of change}$

1 point is earned for the value of v .

2-1-1

The average rate of change of W over the interval $5 \leq v \leq 60$ is $\frac{W(60) - W(5)}{60 - 5} = -0.253$ or -0.254 .

$$W'(v) = \frac{W(60) - W(5)}{60 - 5} \text{ when } v = 23.011.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v .

The average rate of change of W over the interval $5 \leq v \leq 60$ is $\frac{W(60) - W(5)}{60 - 5} = -0.253$ or -0.254 .

$$W'(v) = \frac{W(60) - W(5)}{60 - 5} \text{ when } v = 23.011.$$

Part C

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

$$\left. \frac{dW}{dt} \right|_{t=3} = \left(\frac{dW}{dv} \cdot \frac{dv}{dt} \right) \Big|_{t=3} = W'(35) \cdot 5 = -0.892^\circ\text{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$$\left. \frac{dW}{dt} \right|_{t=3} = -0.892^\circ\text{F/hr}$$

1 point is earned for units in (a) and (c)

2-1-1

Units of °F/mph in (a) and °F/hr in (c)



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

$$\left. \frac{dW}{dt} \right|_{t=3} = \left(\frac{dW}{dv} \cdot \frac{dv}{dt} \right) \bigg|_{t=3} = W'(35) \cdot 5 = -0.892^\circ\text{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$$\left. \frac{dW}{dt} \right|_{t=3} = -0.892^\circ\text{F/hr}$$

1 point is earned for units in (a) and (c)

Units of °F/mph in (a) and °F/hr in (c)